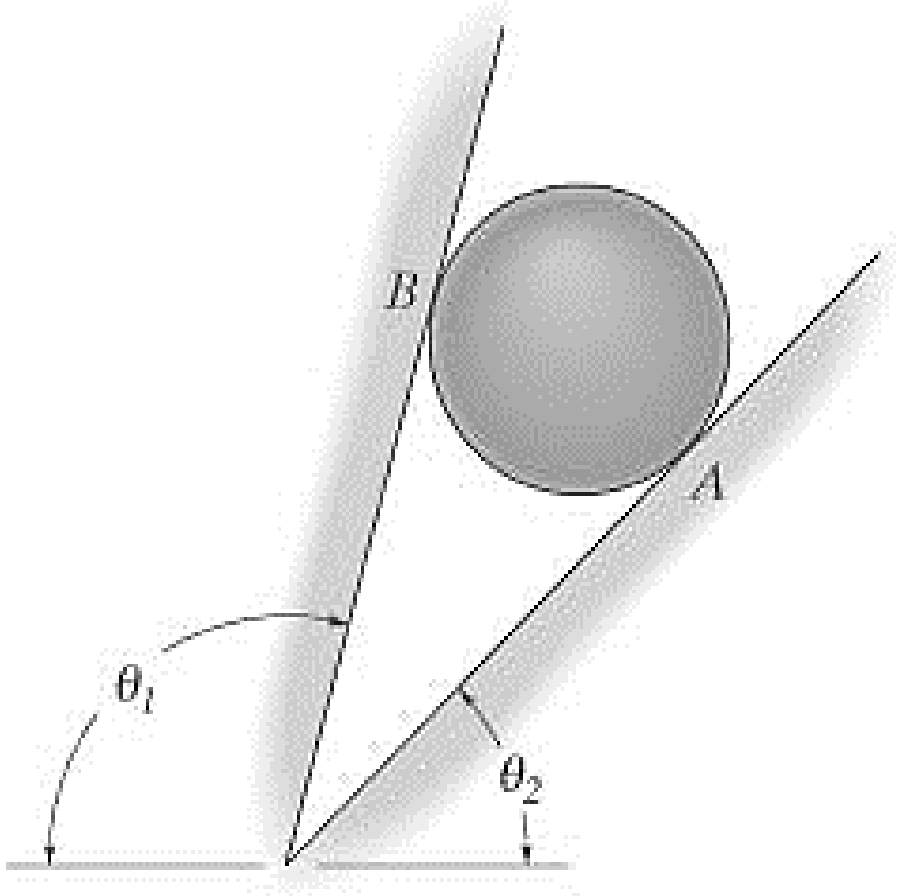
# STATİK
# SERBEST CİSİM DİYAGRAMLARI-2D

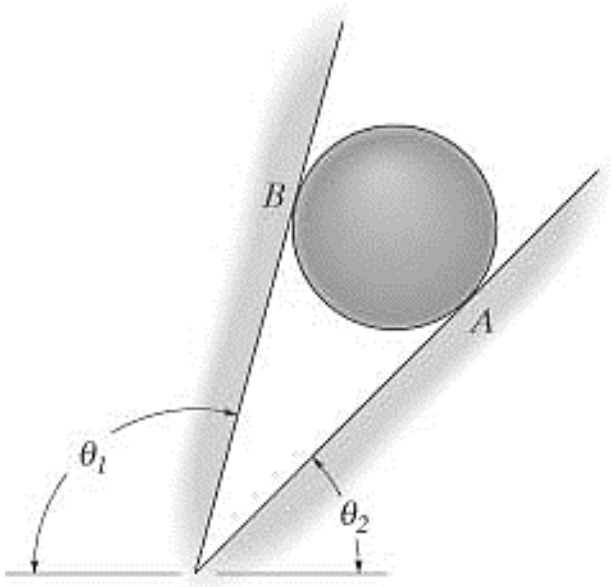
150 kN ağırlığındaki şekildeki küre eğik ve pürüzsüz iki yüzey arasında şekilde gösterildiği gibi dengededir. Serbest cisim diyagramını çizerek A ve B deki tepki kuvvetlerini hesaplayınız.

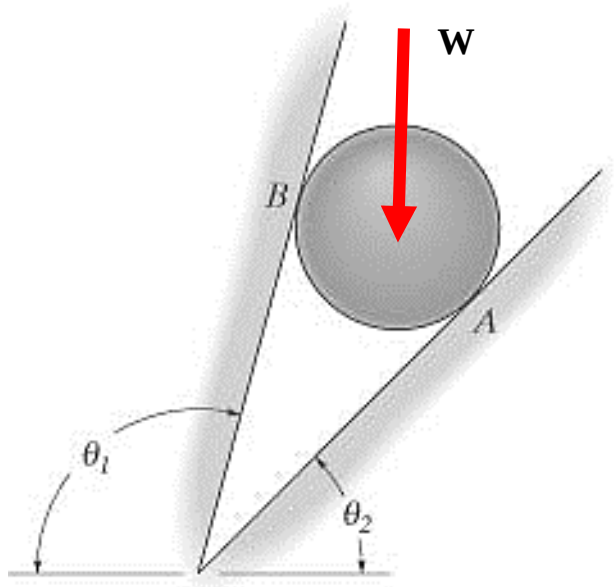
$\theta_1=105°$ ,$\theta_2=45°$



150 kN ağırlığındaki şekildeki küre eğik ve pürüzsüz iki yüzey arasında şekilde gösterildiği gibi dengededir. Serbest cisim diyagramını çizerek A ve B deki tepki kuvvetlerini hesaplayınız.

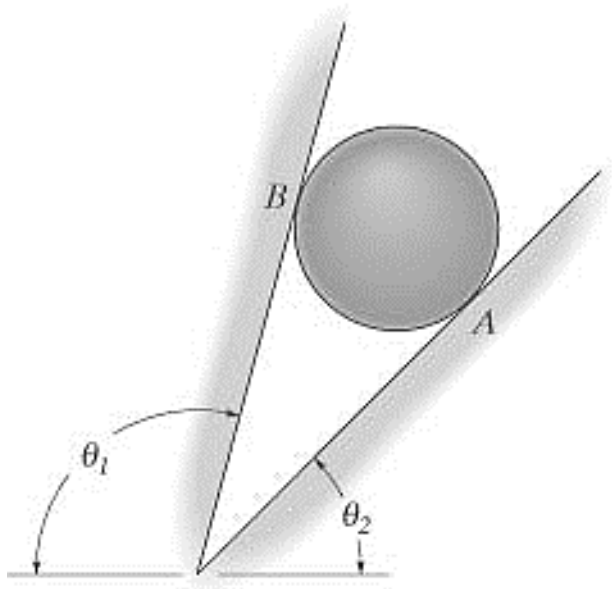
$\theta_1=105°$ ,$\theta_2=45°$

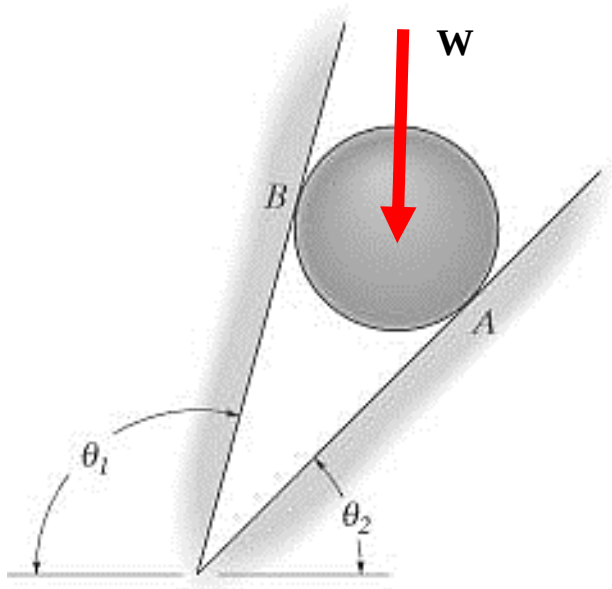


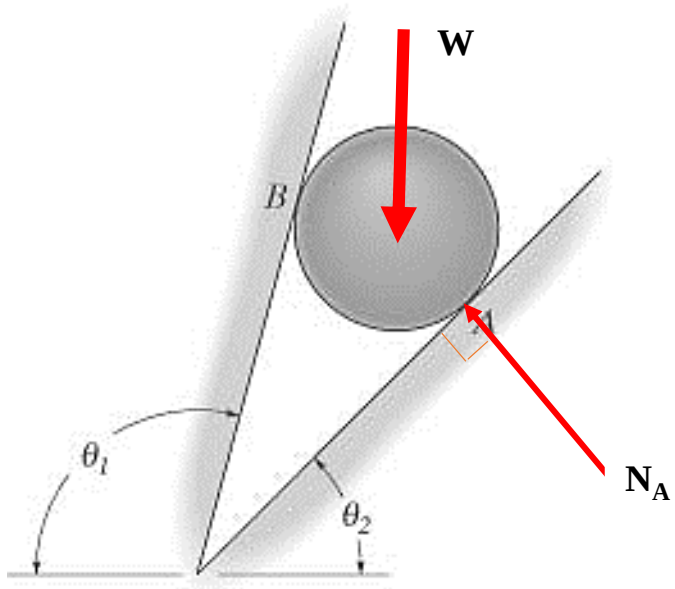



150 kN ağırlığındaki şekildeki küre eğik ve pürüzsüz iki yüzey arasında şekilde gösterildiği gibi dengededir. Serbest cisim diyagramını çizerek A ve B deki tepki kuvvetlerini hesaplayınız.

$\theta_1=105°$ ,$\theta_2=45°$

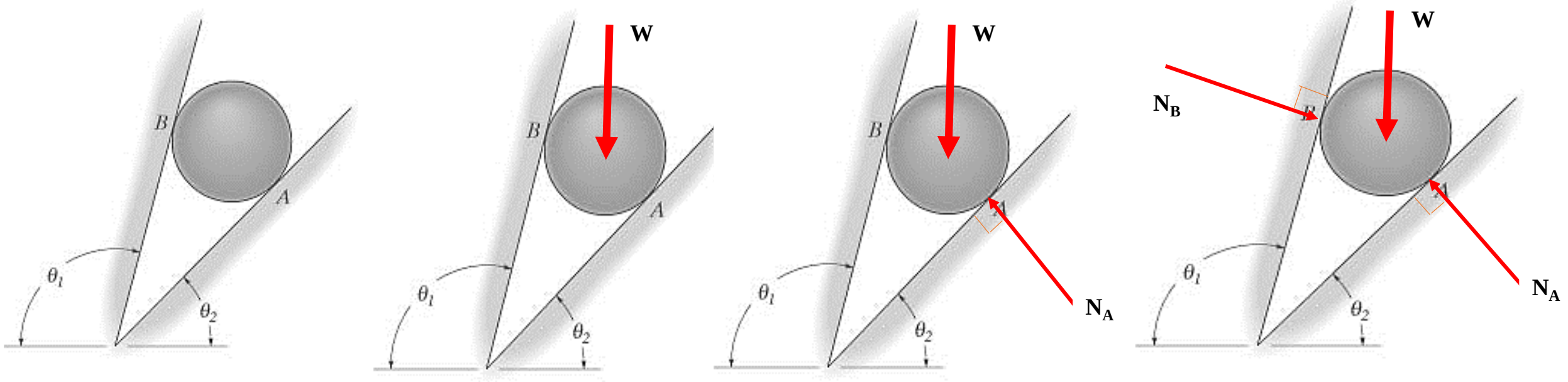


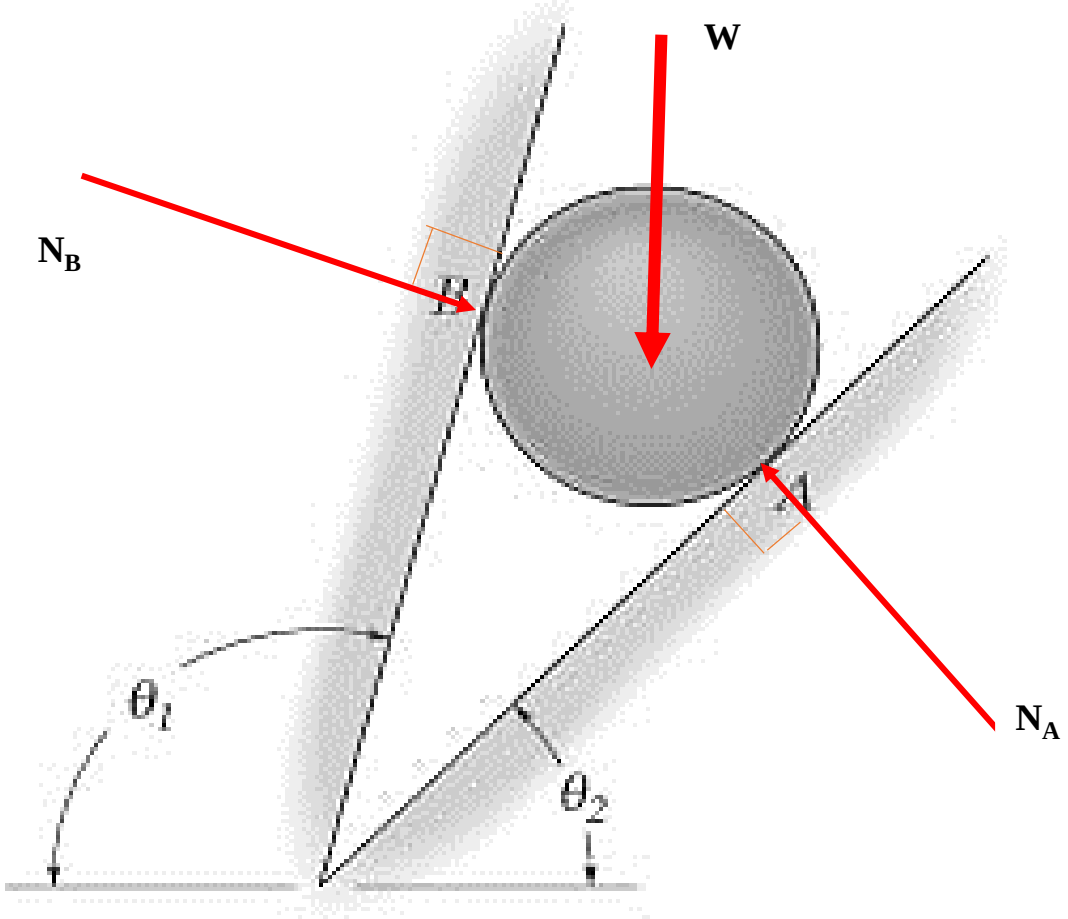
150 kN ağırlığındaki şekildeki küre eğik ve pürüzsüz iki yüzey arasında şekilde gösterildiği gibi dengededir. Serbest cisim diyagramını çizerek A ve B deki tepki kuvvetlerini hesaplayınız.

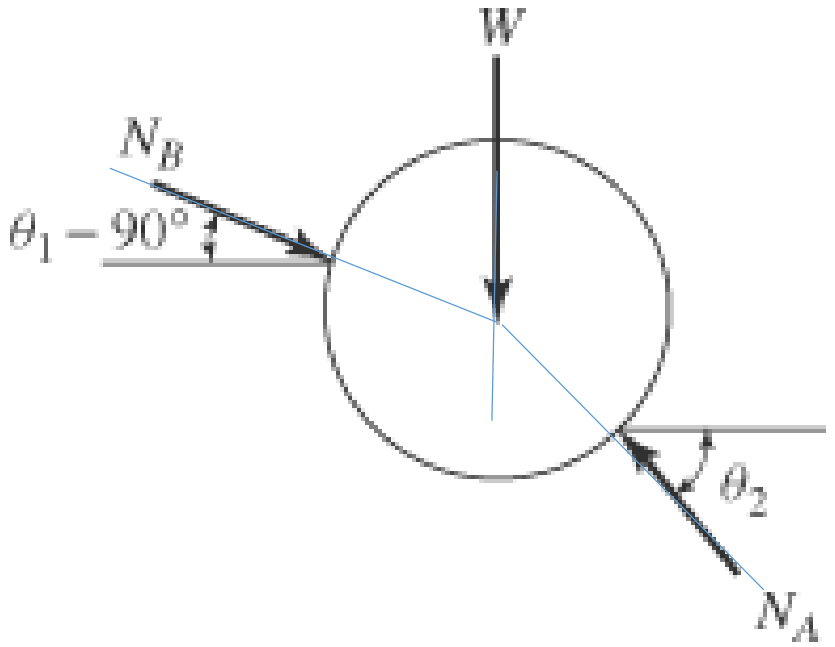
$\theta_1=105°$ ,$\theta_2=45°$



150 kN ağırlığındaki şekildeki küre eğik ve pürüzsüz iki yüzey arasında şekilde gösterildiği gibi dengededir. Serbest cisim diyagramını çizerek A ve B deki tepki kuvvetlerini hesaplayınız.

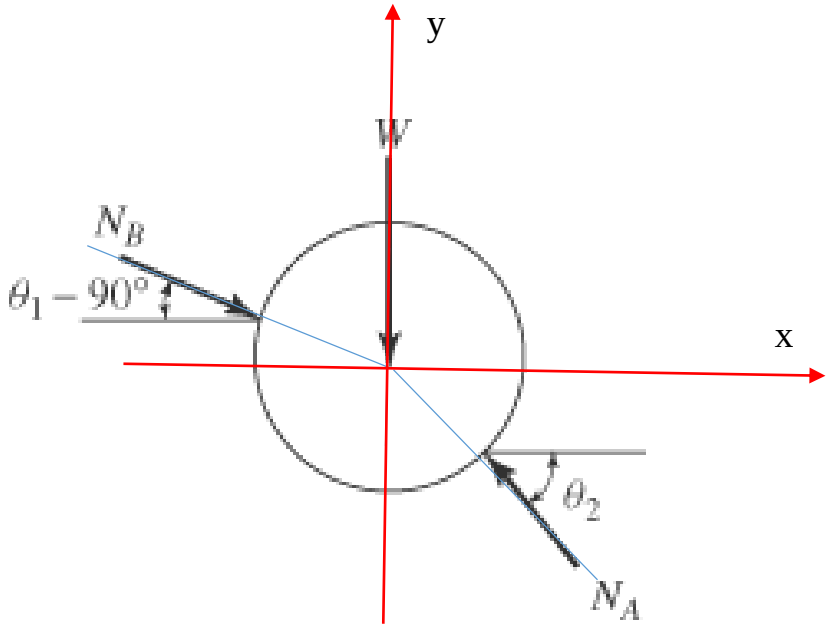
$\theta_1=105°$ ,$\theta_2=45°$



150 kN ağırlığındaki şekildeki küre eğik ve pürüzsüz iki yüzey arasında şekilde gösterildiği gibi dengededir. Serbest cisim diyagramını çizerek A ve B deki tepki kuvvetlerini hesaplayınız.

$\theta_1$=105° ,$\theta_2$=45°

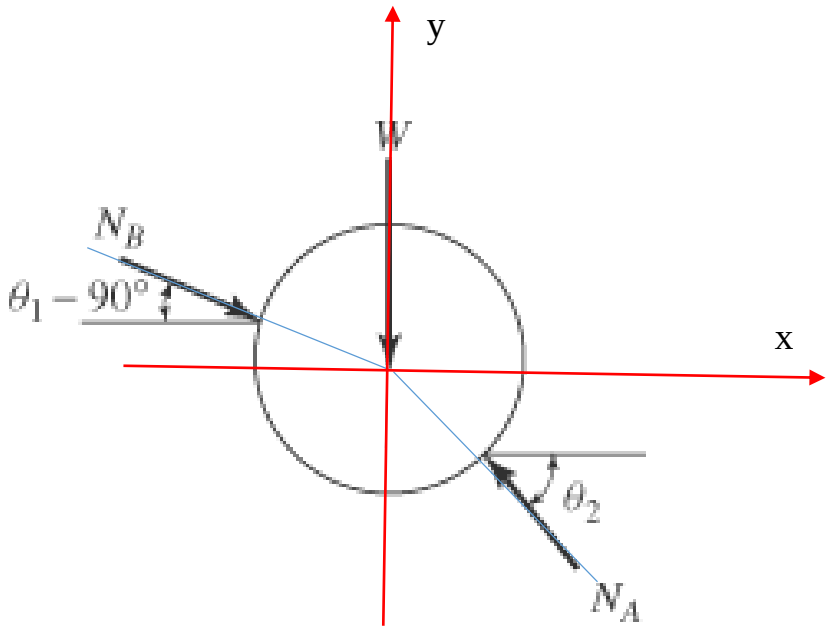



$$\sum F_x = 0 \quad \pm$$

$$N_B \cos 15 - N_A \cos 45 = 0$$

150 kN ağırlığındaki şekildeki küre eğik ve pürüzsüz iki yüzey arasında şekilde gösterildiği gibi dengededir. Serbest cisim diyagramını çizerek A ve B deki tepki kuvvetlerini hesaplayınız.

$\theta_1$=105° ,$\theta_2$=45°




$$\sum F_x = 0 \quad \pm$$

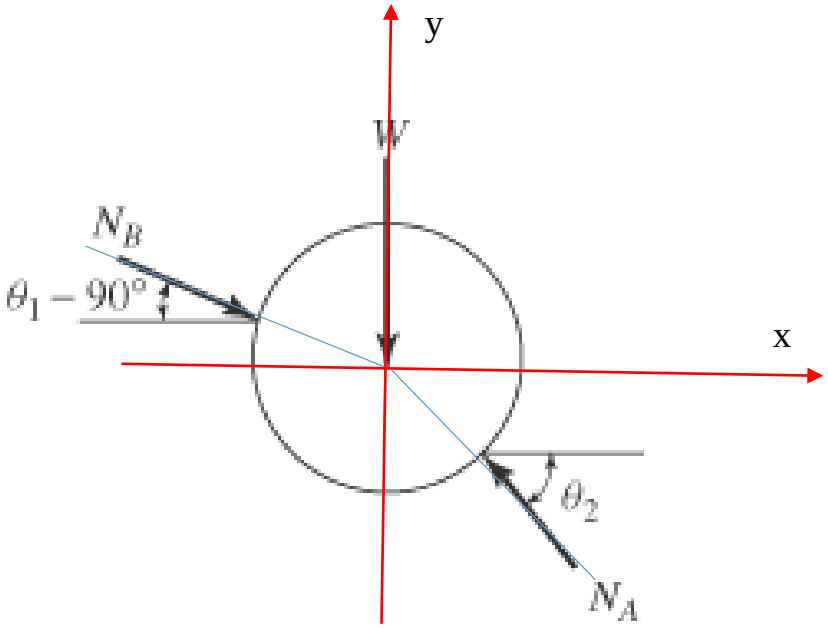
$$N_B \cos 15 - N_A \cos 45 = 0$$

$$\sum F_y = 0 \overset{+}{\uparrow\downarrow}$$

$$\sin 45 N_A - \sin 15 N_B - W = 0$$

$$\sin 45 N_A - \sin 15 N_B = W$$

$$\sin 45 N_A - \sin 15 N_B = 150$$



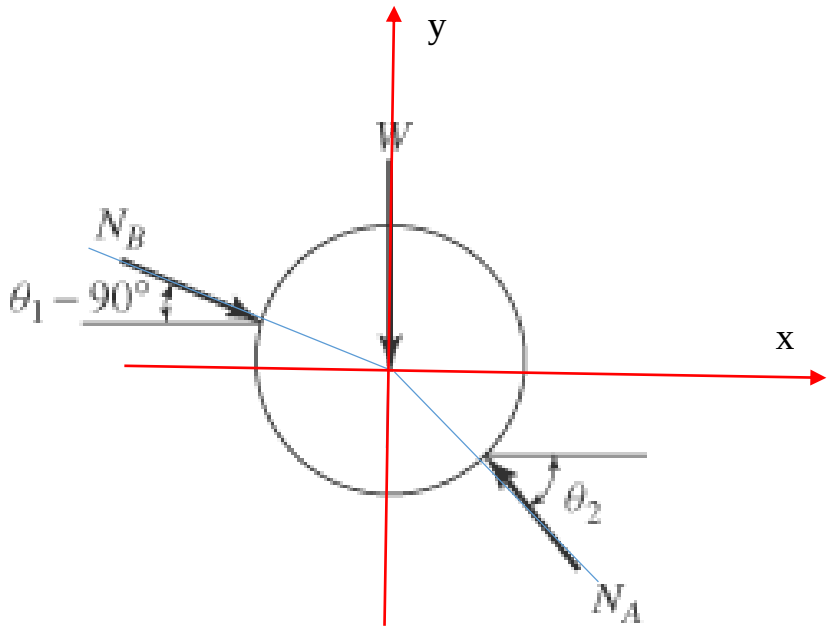

$$N_B \cos 15 - N_A \cos 45 = 0$$

$$+ \quad \sin 45 N_A - \sin 15 N_B = 150$$

$$N_B \cos 15 - N_B \sin 15 = 150$$

$$N_B (\cos 15 - \sin 15) = 150$$

$$N_B = \frac{150}{\cos 15 - \sin 15} = 212.132 kN$$




$$N_B \cos 15 - N_A \cos 45 = 0$$

$$+ \quad \sin 45 N_A - \sin 15 N_B = 150$$

$$N_B \cos 15 - N_B \sin 15 = 150$$

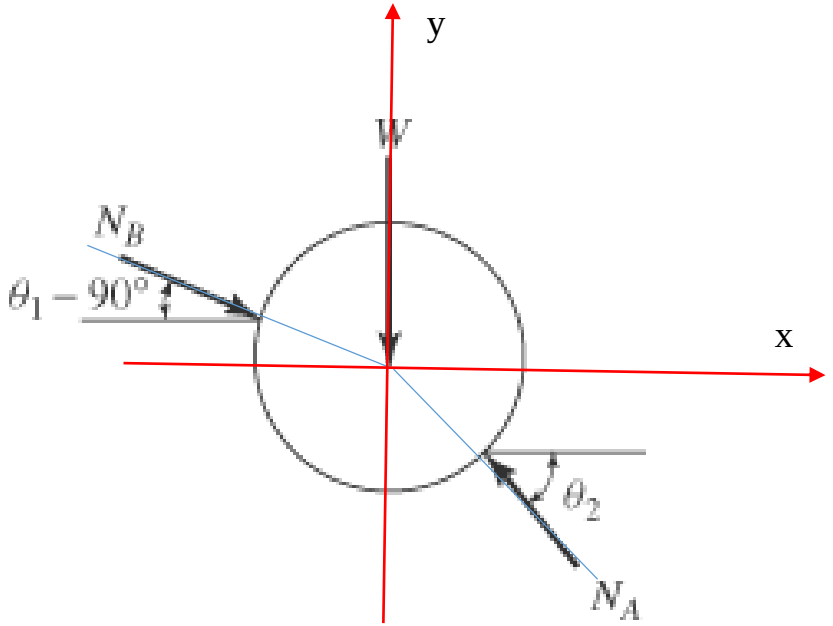
$$N_B (\cos 15 - \sin 15) = 150$$

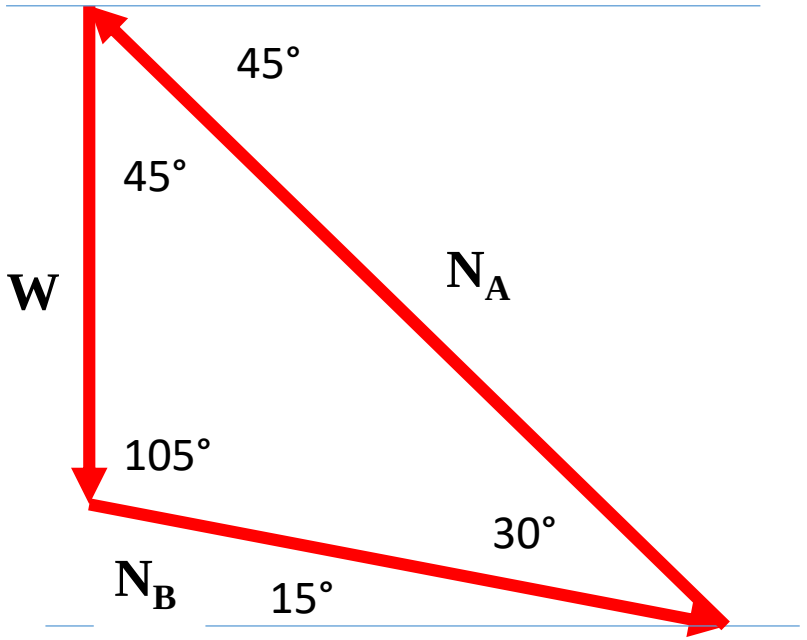
$$N_B = \frac{150}{\cos 15 - \sin 15} = 212.132 kN$$

$$N_B \cos 15 - N_A \cos 45 = 0$$

$$212.132 \cos 15 - N_A \cos 45 = 0$$

$$N_A = \frac{212.132 \cos 15}{\cos 45} = 289.77 kN$$



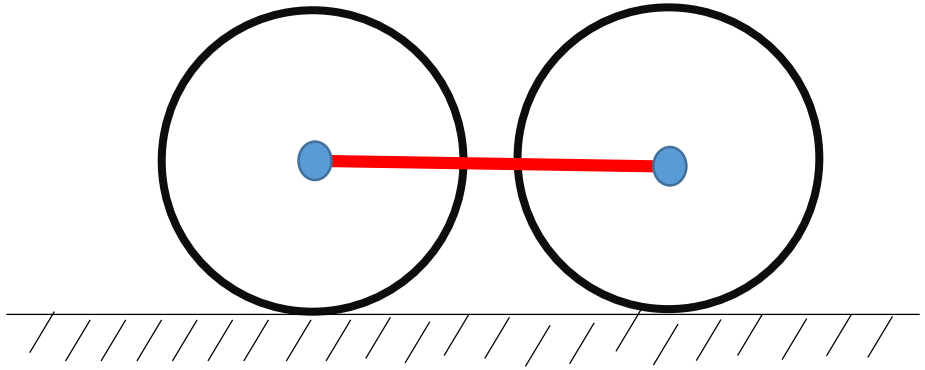



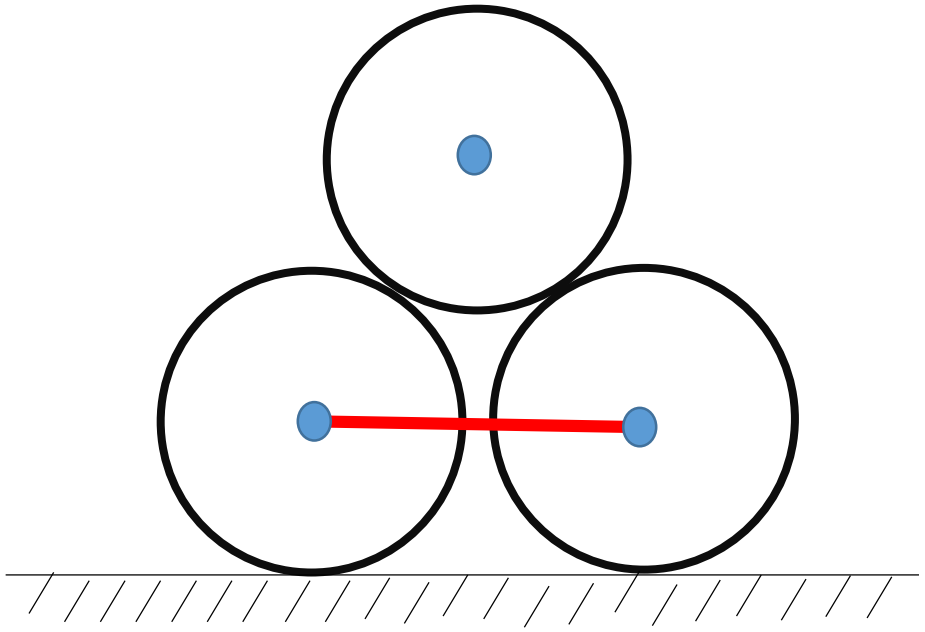

$$\frac{W}{\sin 30} = \frac{N_A}{\sin 105} = \frac{N_B}{\sin 45}$$

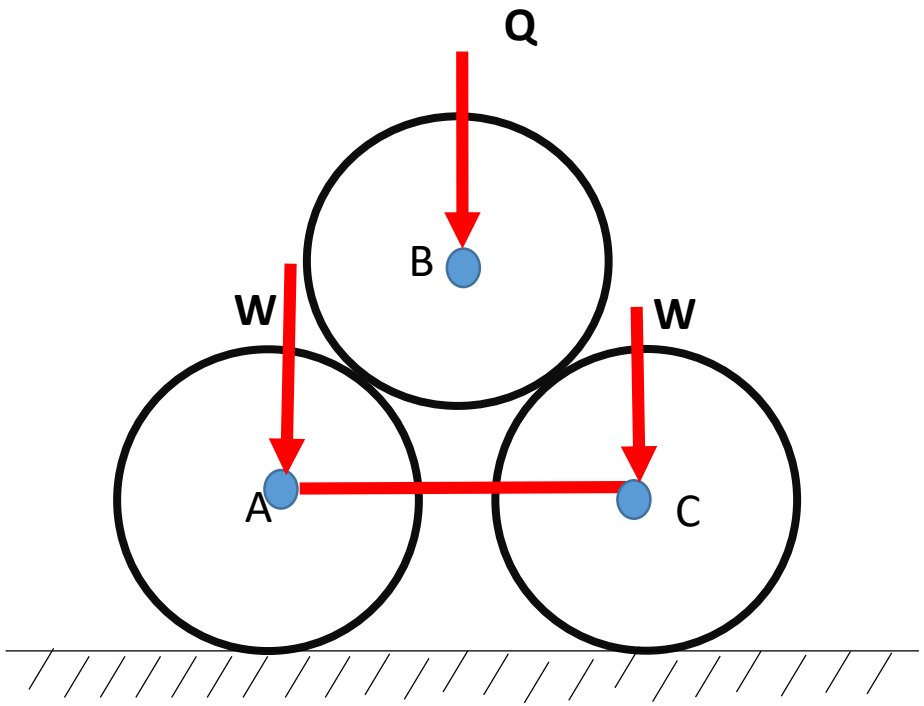
$$N_A = \frac{W \sin 105}{\sin 30} = \frac{150 \sin 105}{\sin 30} = 289.77 kN$$

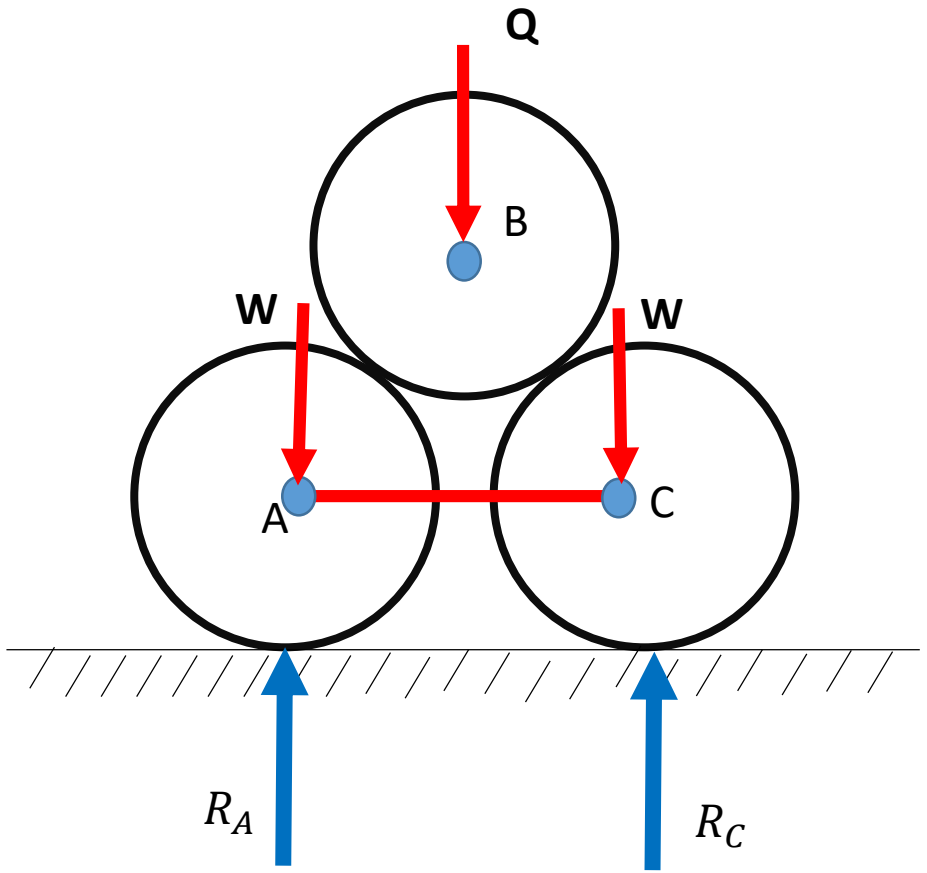
$$N_B = \frac{W \sin 45}{\sin 30} = \frac{150 \sin 45}{\sin 30} = 212.132 kN$$

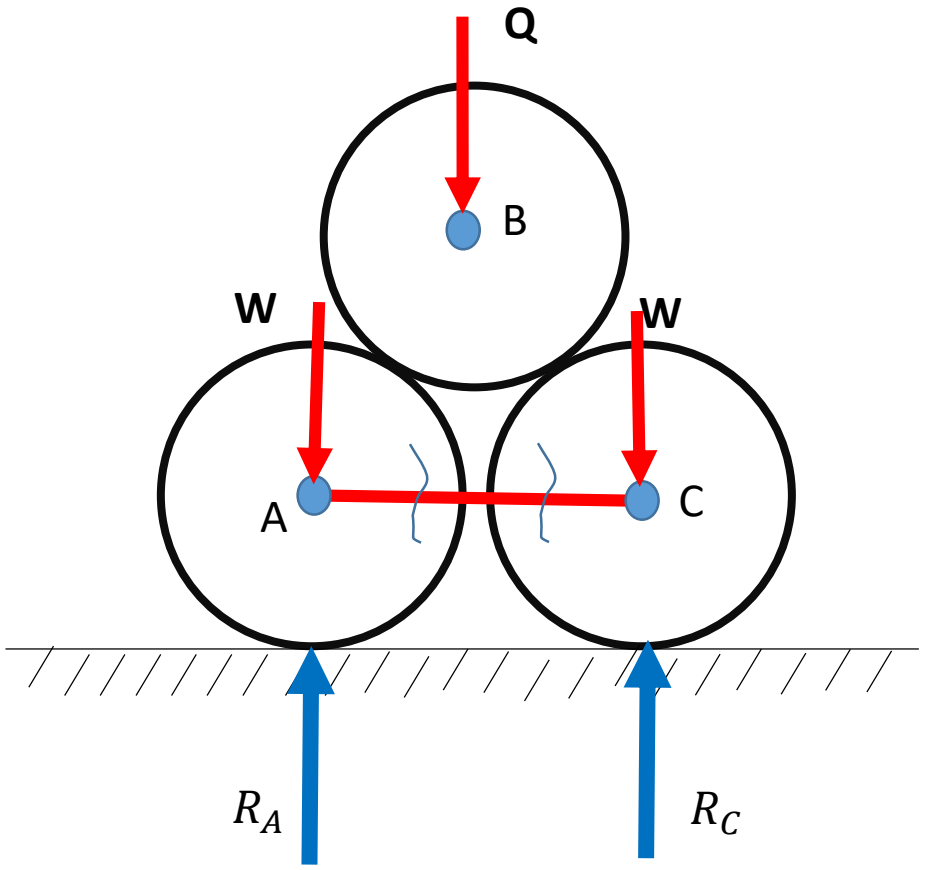
W=445 N ağırlığında ve yarıçapları 152 mm olan 2 silindir pürüzsüz yatay zemin üzerinde merkezleri arasında 406 mm mesafe bulunduğunda merkezlerinden AB halatı ile bağlanmıştır. Bu esnada aynı yarıçapta Q= 890 N olan üçüncü silindir şekilde gösterildiği gibi yerleştirilmiştir. Serbest cisim diyagramını çizerek A ve B deki tepki kuvvetlerini, halattaki çekme kuvvetini hesaplayınız.

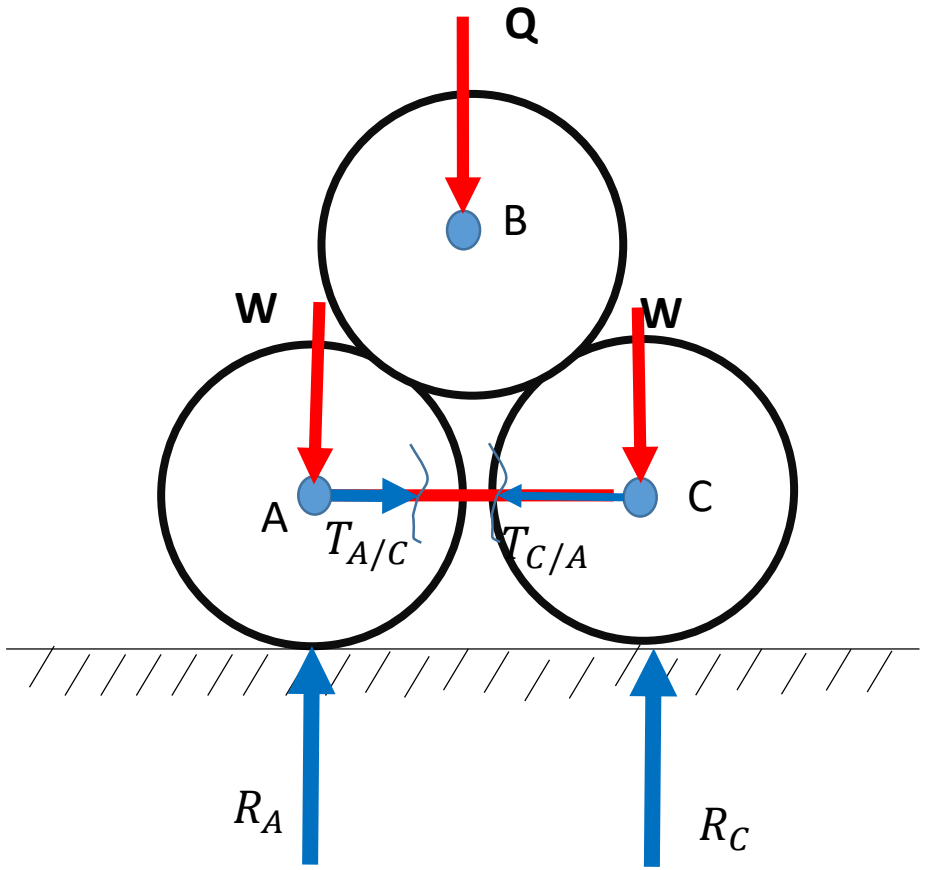




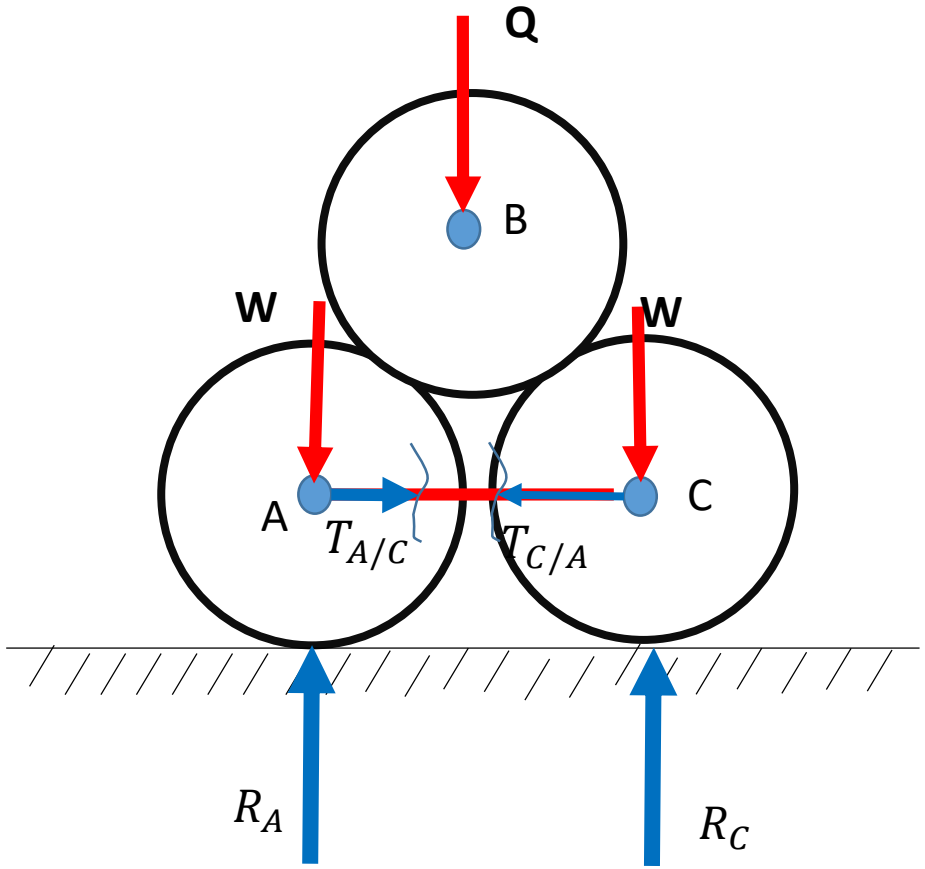

Q
B
W
W
A
C

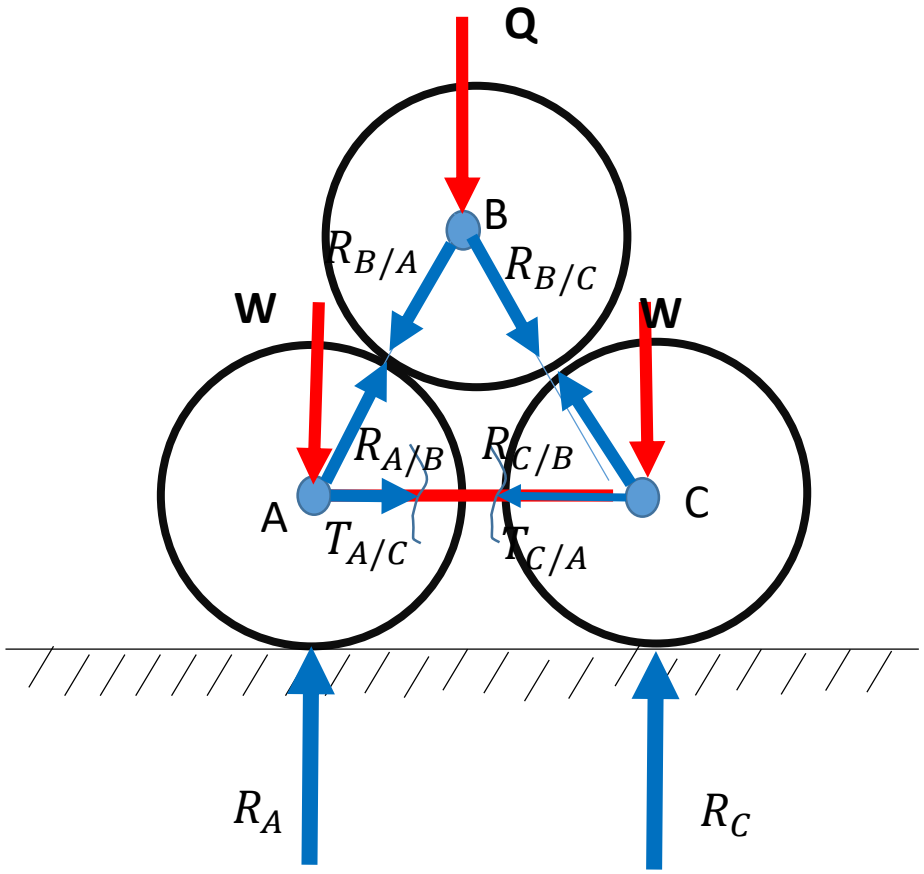

Q
B
W
W
A
C
$R_A$
$R_C$




$$T_{A/C} = T_{C/A}$$

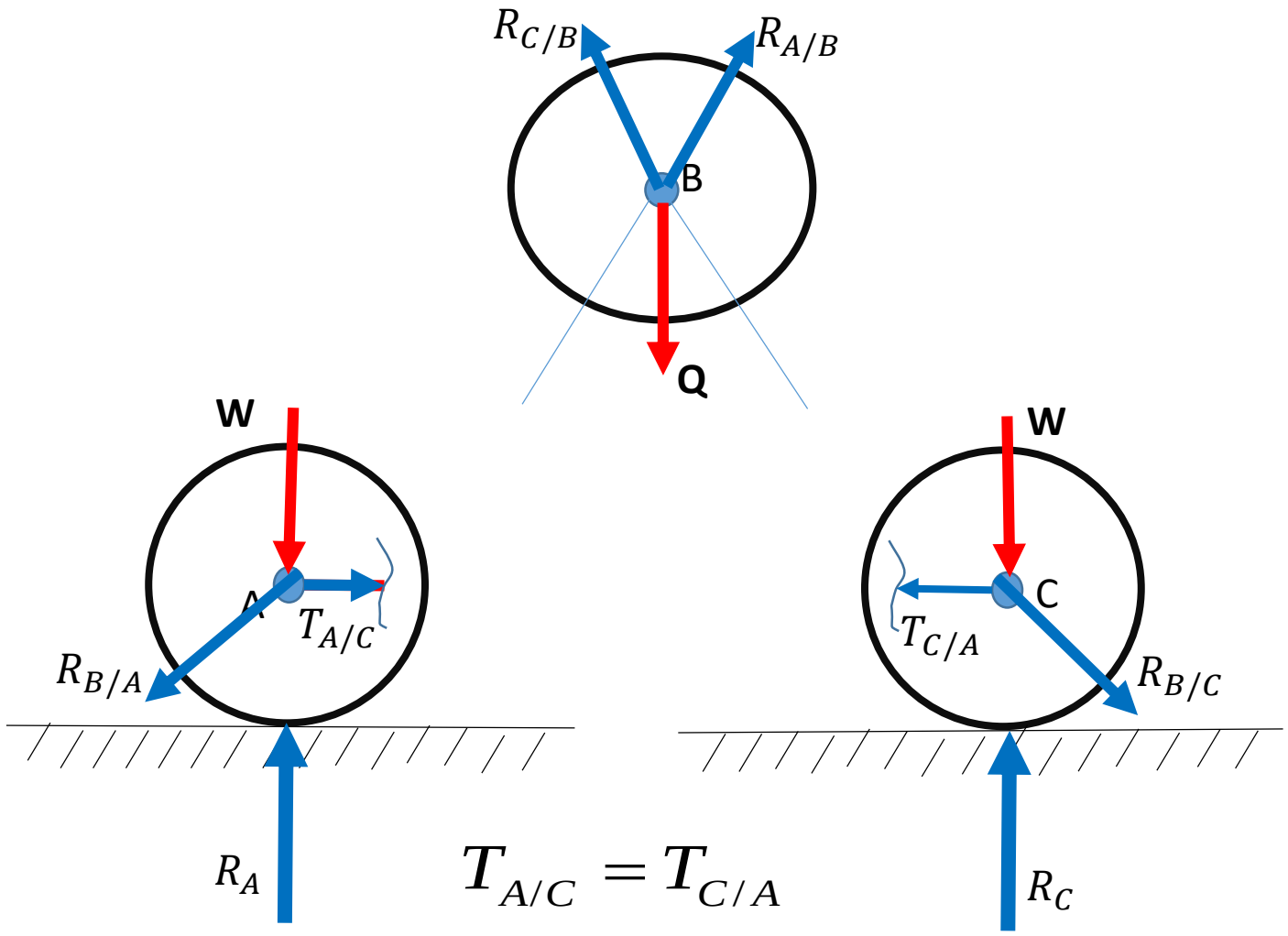



$$T_{A/C} = T_{C/A}$$

$$R_{A/B} = R_{B/A}$$

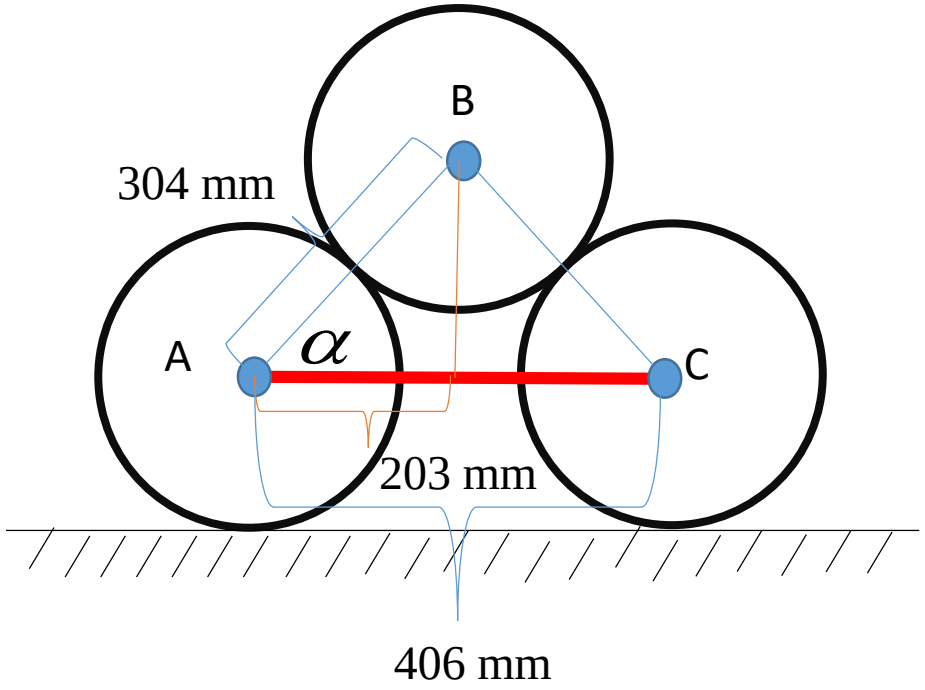
$$R_{B/C} = R_{C/B}$$




$$T_{A/C} = T_{C/A}$$

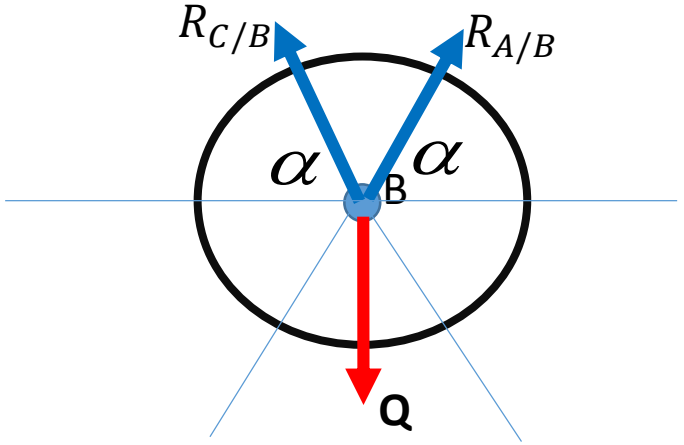
$$R_{A/B} = R_{B/A}$$

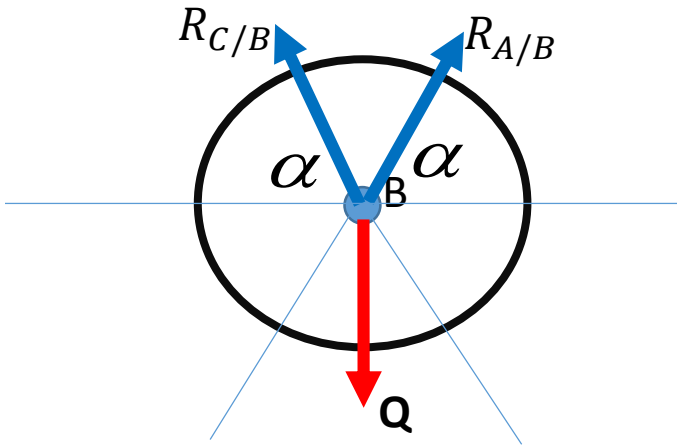
$$R_{B/C} = R_{C/B}$$




$$\cos\alpha = \frac{203}{304}$$

$$\alpha = 48.1°$$

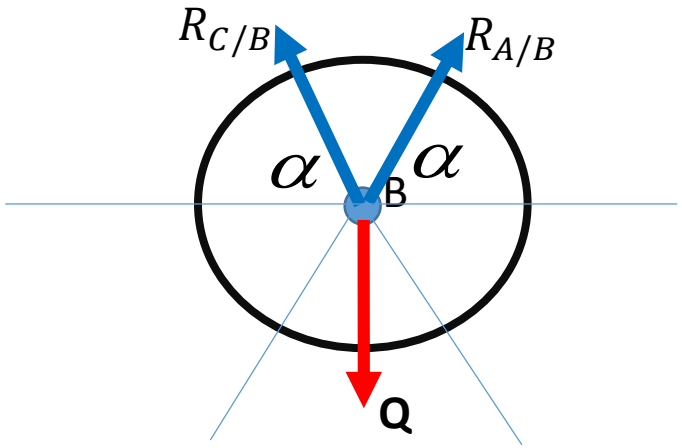





$$\sum F_x = 0 \square \quad \pm$$

$$-R_{C/B}\cos 48.1 + R_{A/B}\cos 48.1 = 0$$

$$R_{C/B} = R_{A/B}$$




$$\sum F_x = 0 \quad \pm$$

$$-R_{C/B}\cos 48.1 + R_{A/B}\cos 48.1 = 0$$

$$R_{C/B} = R_{A/B}$$

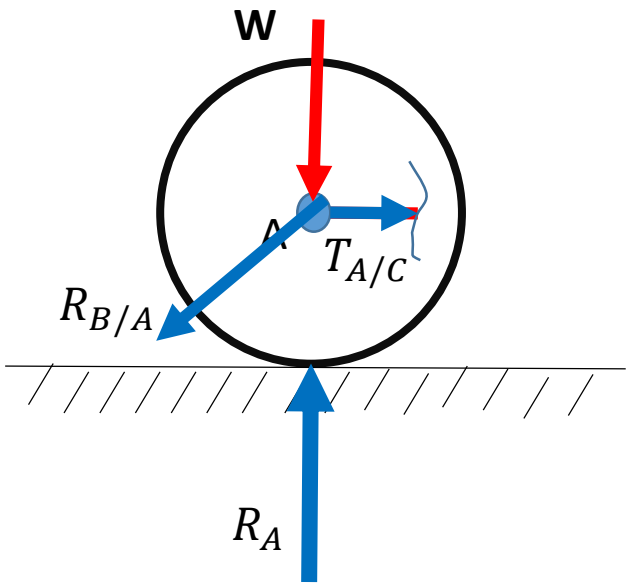
$$\sum F_y = 0 \overset{+}{\uparrow} \underset{-}{\downarrow}$$

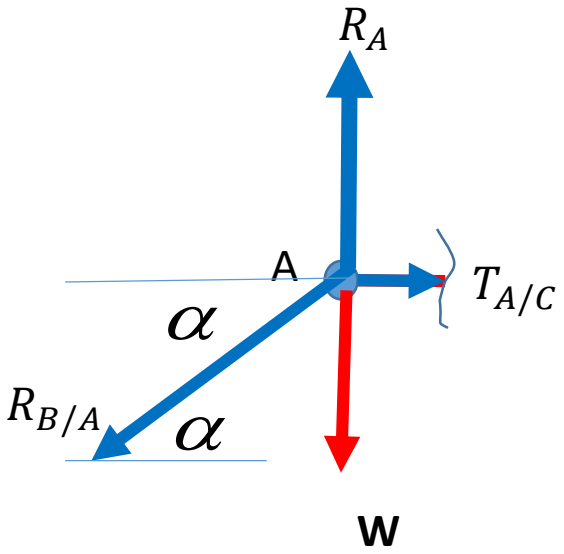
$$\sin 48.1 R_{A/B} + \sin 48.1 R_{C/B} - Q = 0$$

$$\sin 48.1 R_{A/B} + \sin 48.1 R_{C/B} = Q$$

$$\sin 48.1 R_{A/B} + \sin 48.1 R_{C/B} = 890$$

$$R_{A/B} = 597.78N$$




$$\sum F_x = 0 \rightarrow \quad \pm$$

$$-R_{B/A}\cos 48.1 + T_{A/C} = 0$$

$$T_{A/C} = R_{B/A}\cos 48.1$$

$$T_{A/C} = 597.78\cos 48.1$$

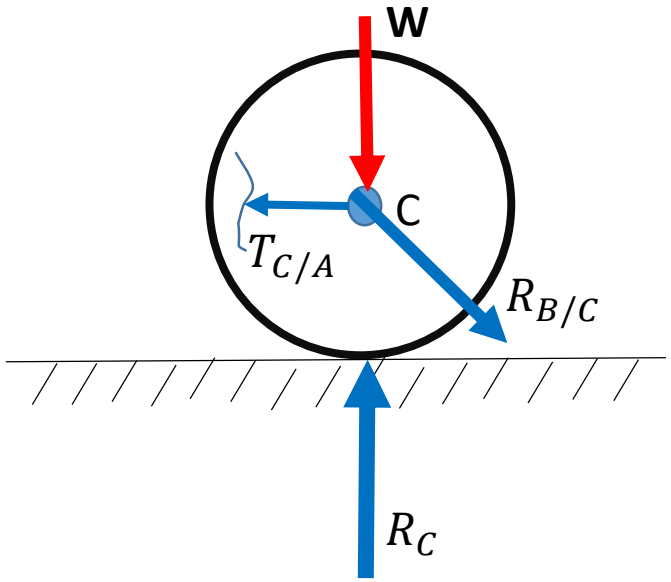
$$T_{A/C} = 399.21N$$

$$\sum F_y = 0 \overset{+}{\uparrow}\underset{-}{\downarrow}$$

$$R_A - \sin 48.1 R_{B/A} - w = 0$$

$$R_A = \sin 48.1 x 597.78 + 445$$

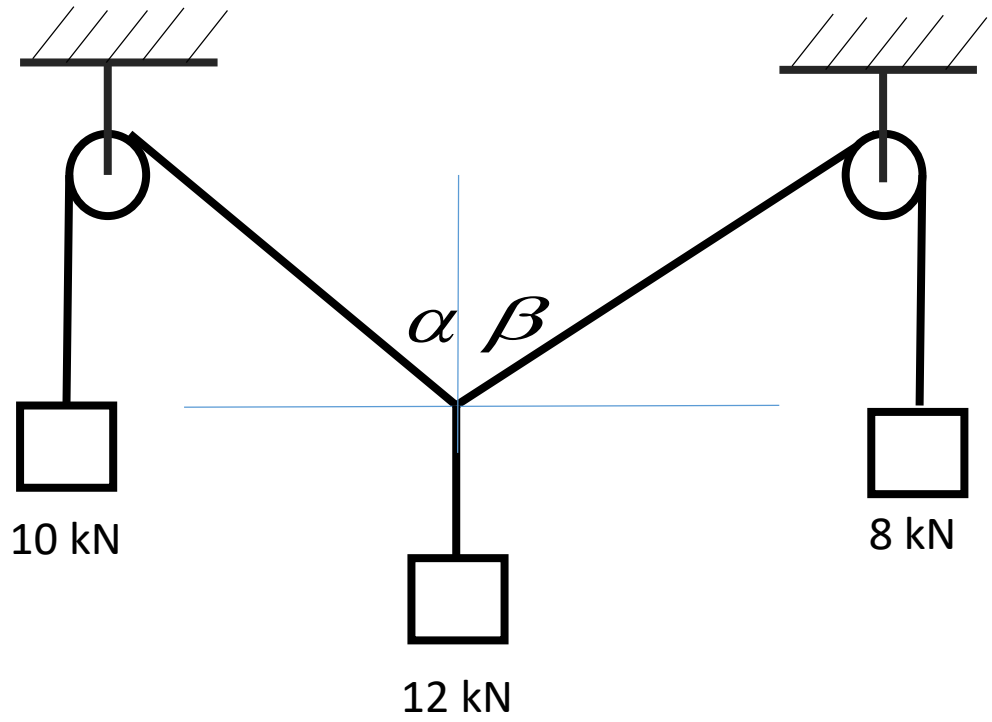
$$R_A = 889.93N$$

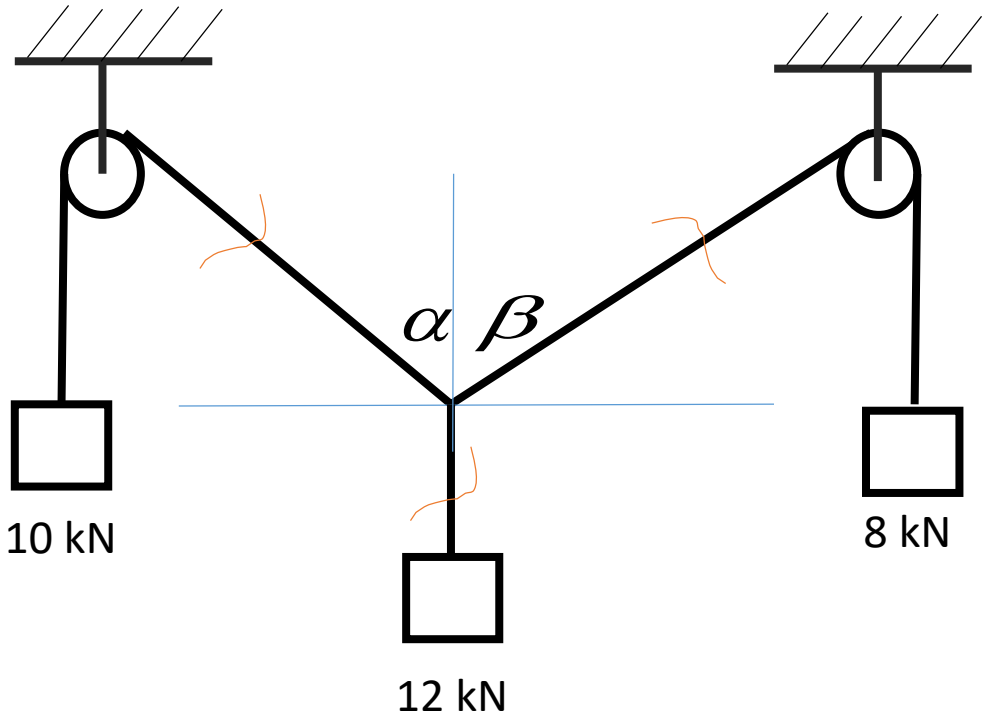


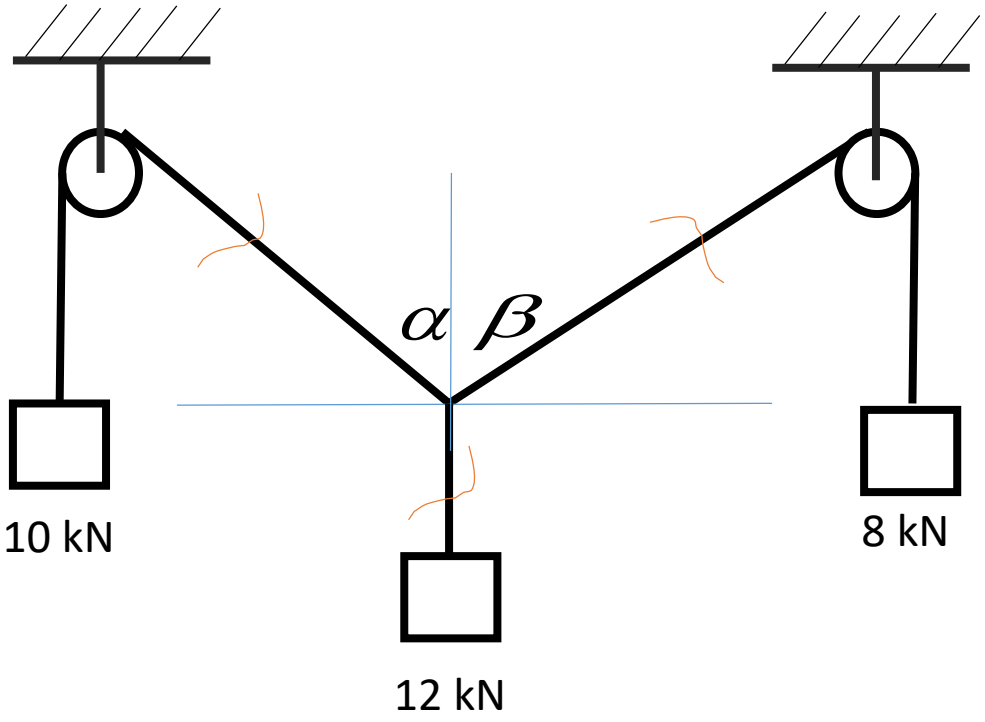

$$R_{C/B} = R_{A/B} = 597.78N$$

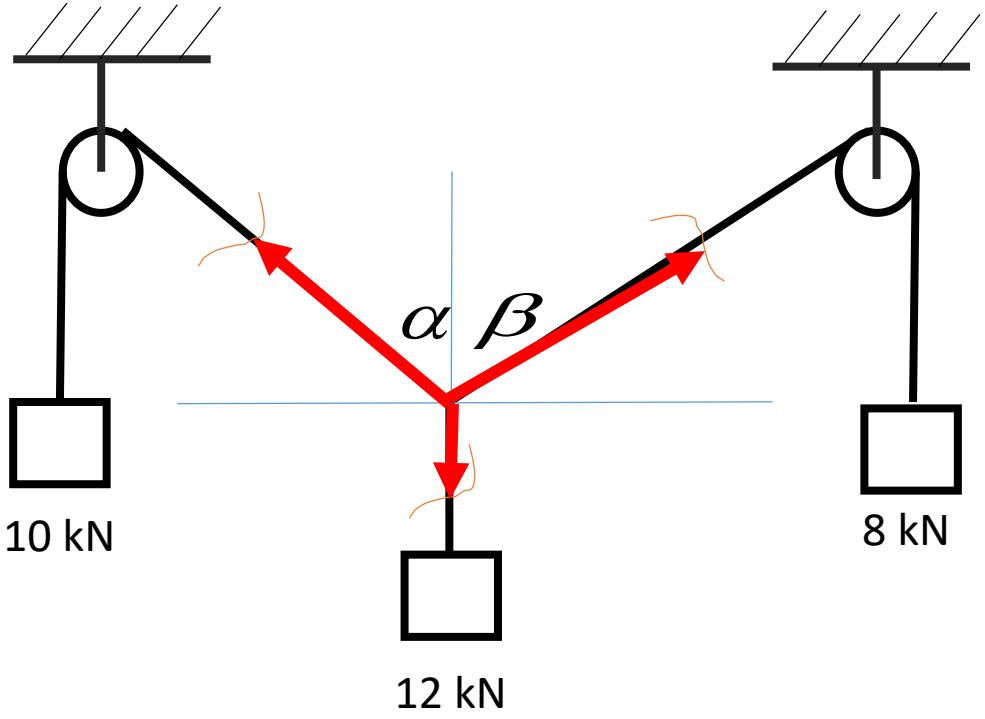
$$R_A = R_C = 889.93N$$

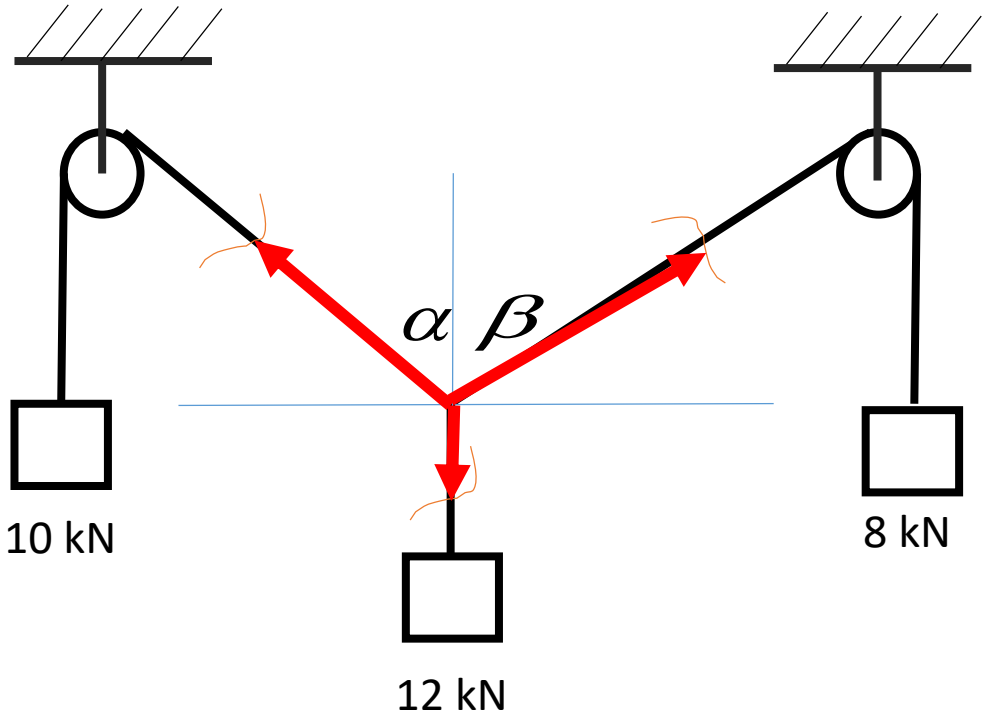
Şekildeki yükler birbirine halatla bağlanmış ve dengededir. Serbest cisim diyagramını çizerek α ve β açılarını hesaplayınız.

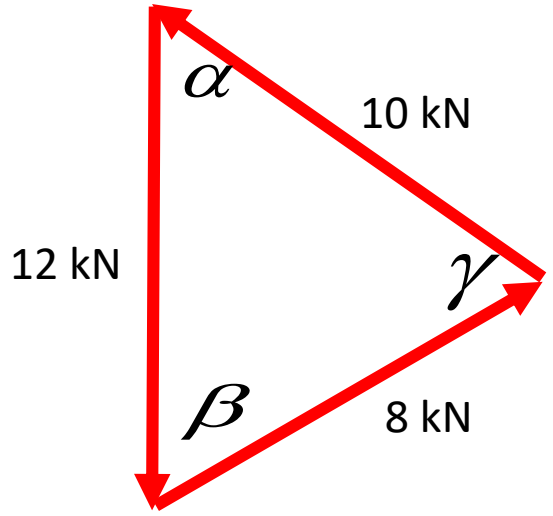



α β
10 kN
12 kN
8 kN


α β
10 kN
8 kN
12 kN


α β
10 kN
8 kN
12 kN



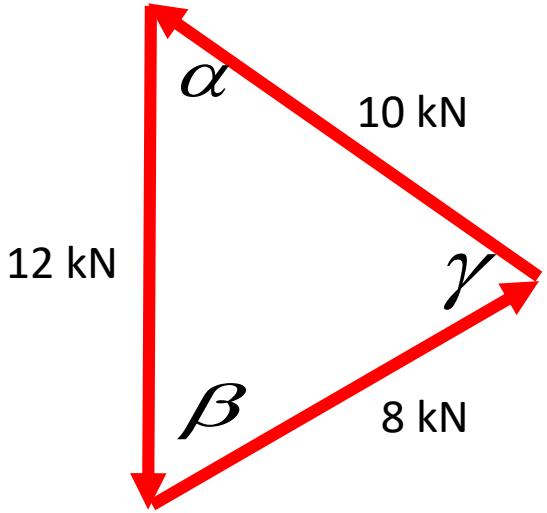




## Kosinüs Teoremi

$$10^2 = 8^2 + 12^2 - 2x8x12x\cos\beta$$

$$\cos\beta = 0.5625$$

$$\beta = 55.771°$$




## Kosinüs Teoremi

$$10^2 = 8^2 + 12^2 - 2x8x12x\cos\beta$$

$$\cos\beta = 0.5625$$

$$\beta = 55.771°$$

## Sinüs Teoremi

$$\frac{10}{\sin 55.77} = \frac{8}{\sin\alpha} = \frac{12}{\sin\gamma}$$

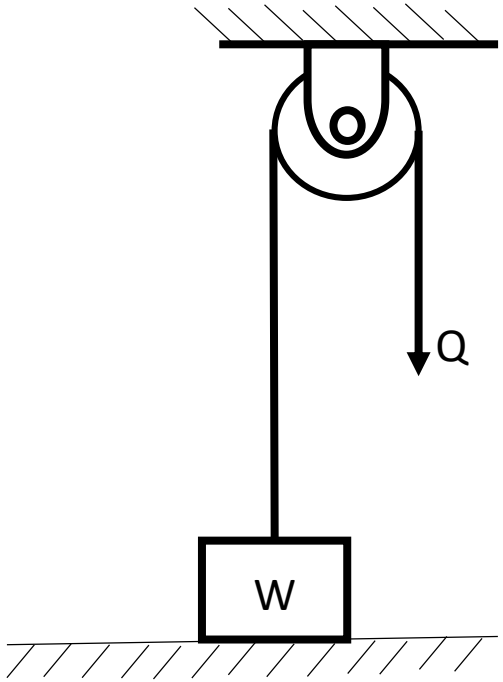
$$\sin\alpha = 0.661$$

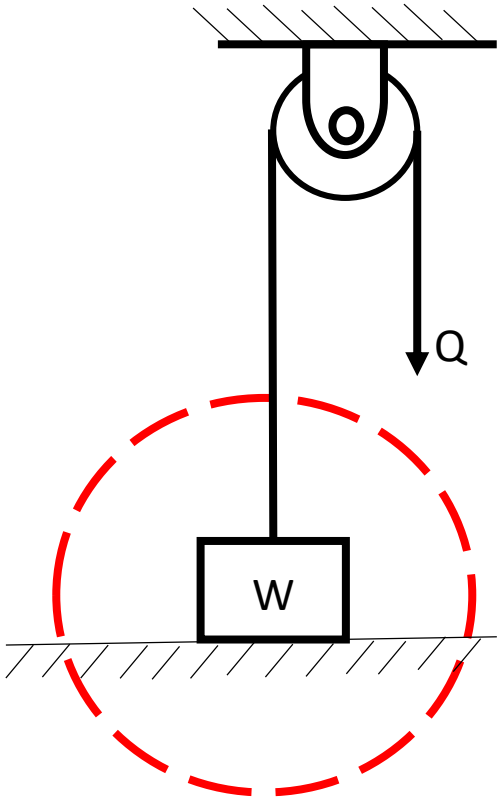
$$\alpha = 41.40°$$

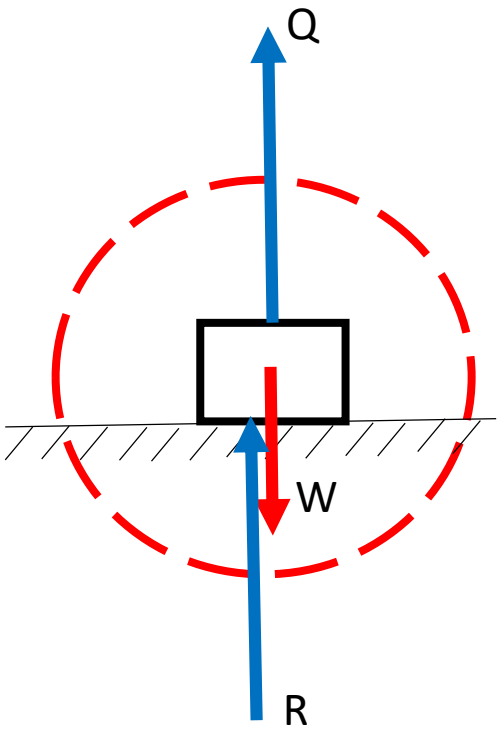
$$\sin\gamma = 0.9921$$

$$\gamma = 82.81°$$

W=712 N bir blok şekilde görüldüğü gibi yatay zemin üzerinde durmaktadır. Blok a bir halat bağlanmış ve sürtünmesiz bir makaradan geçirildikten sonra ucuna Q=534 N luk kuvvet uygulanmaktadır. W bloku ile zemin arasındaki tepki kuvvetini hesaplayınız.





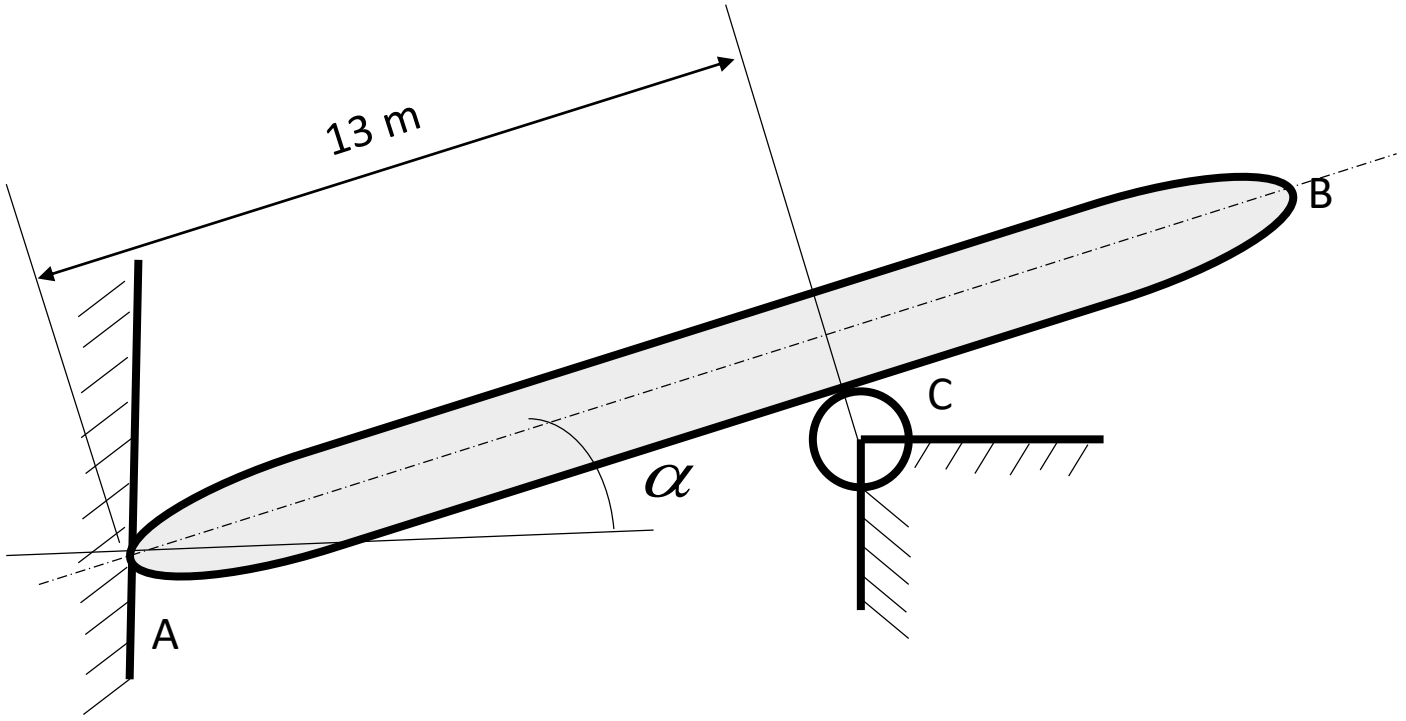

$$\sum F_y = 0 \overset{+}{\uparrow\downarrow}$$

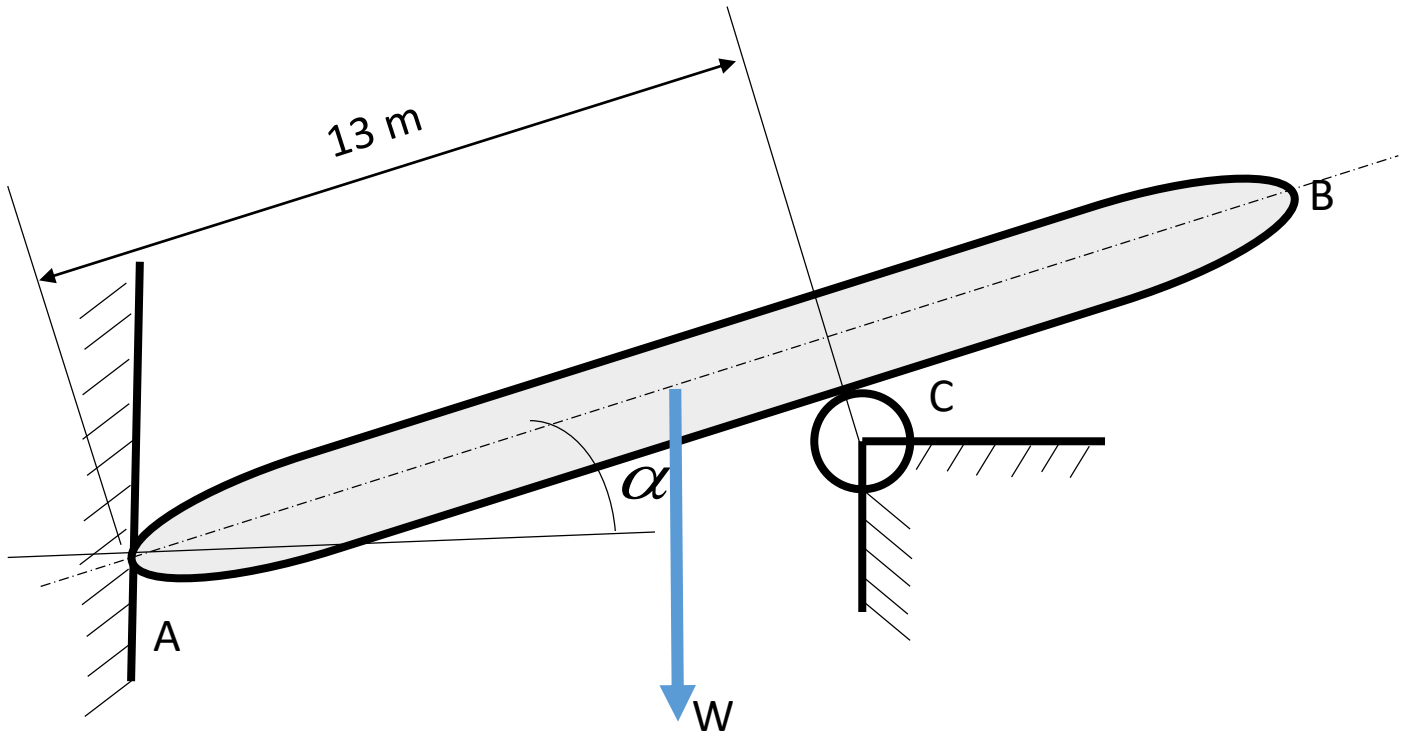
$$Q + R - W = 0$$

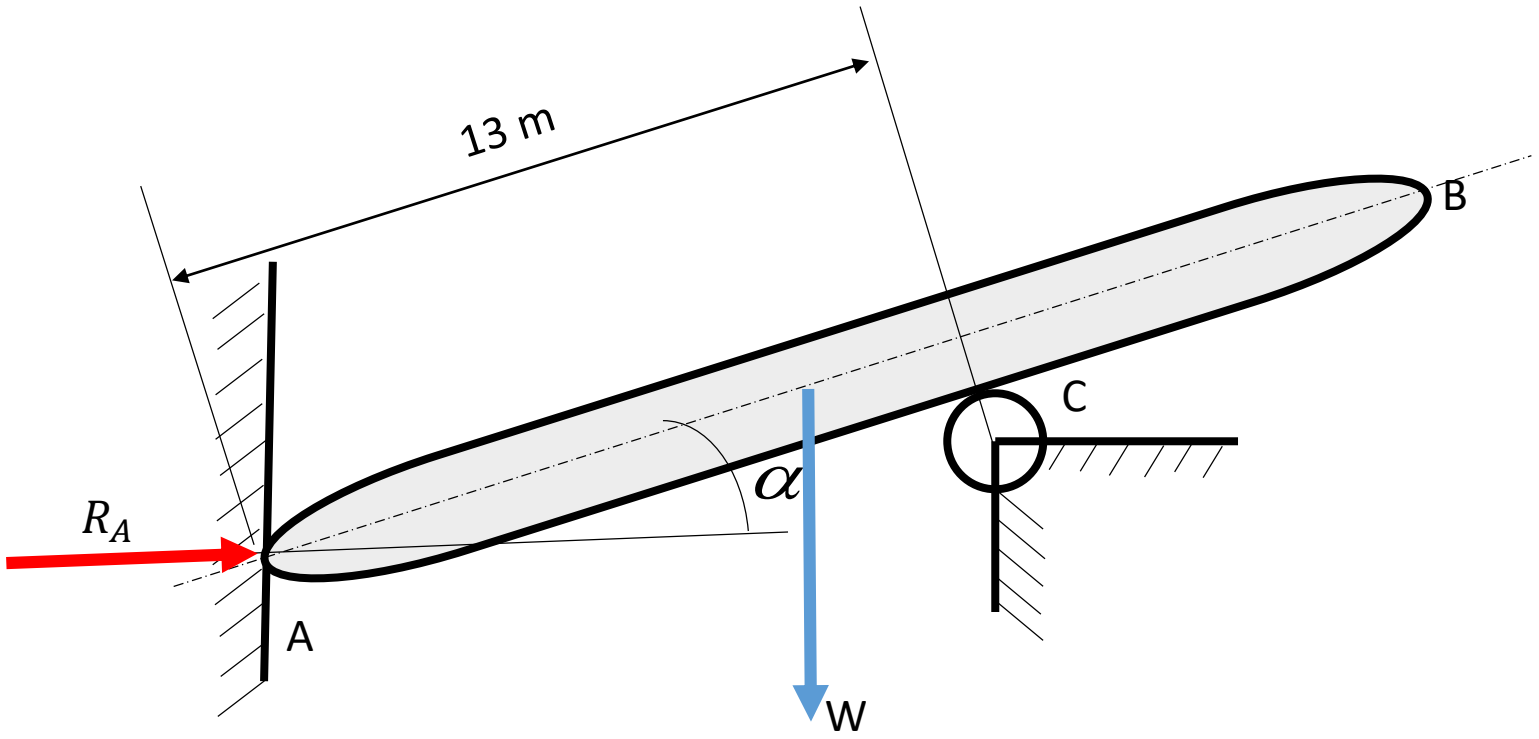
$$534 + R - 712 = 0$$

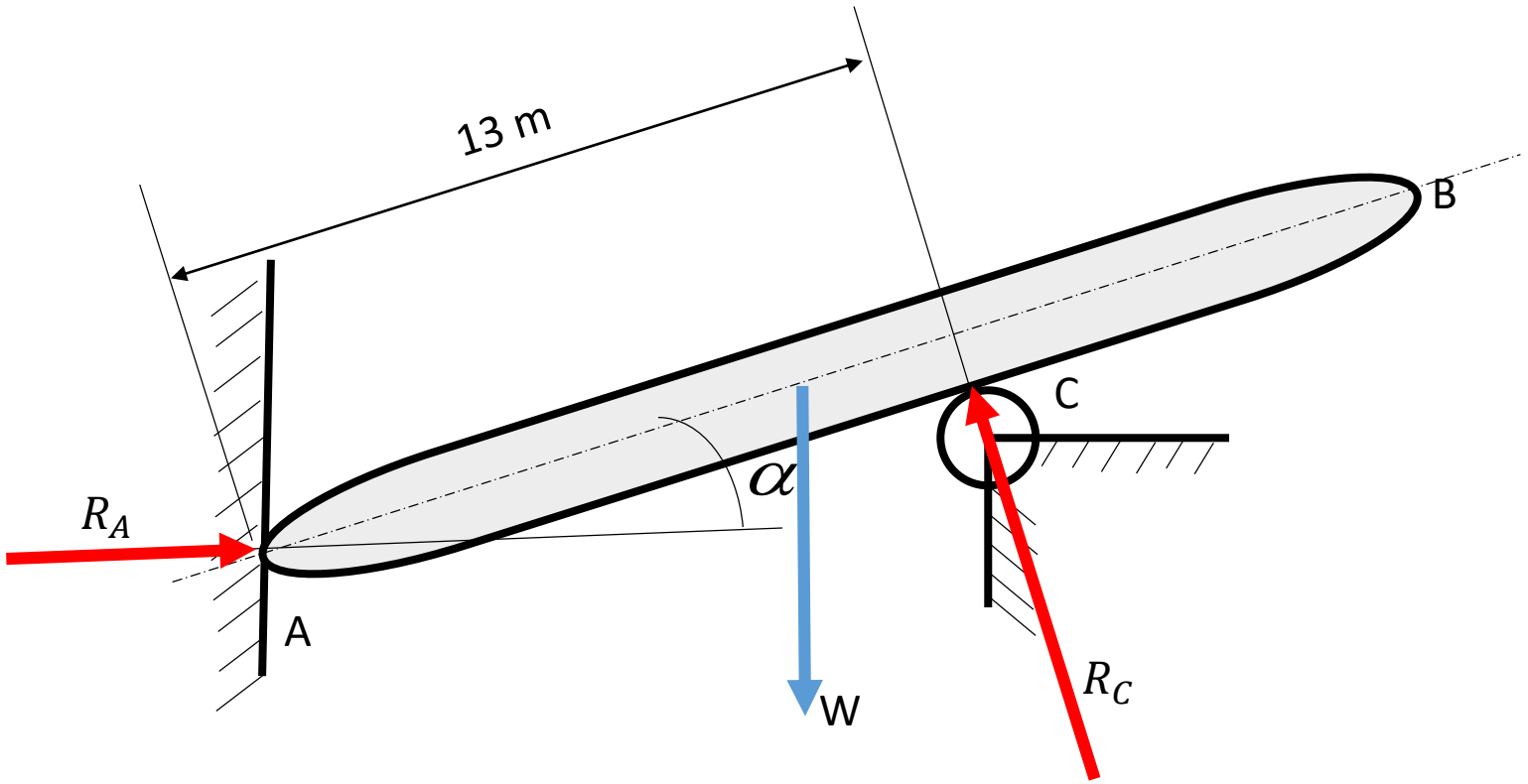
$$R = 178N$$

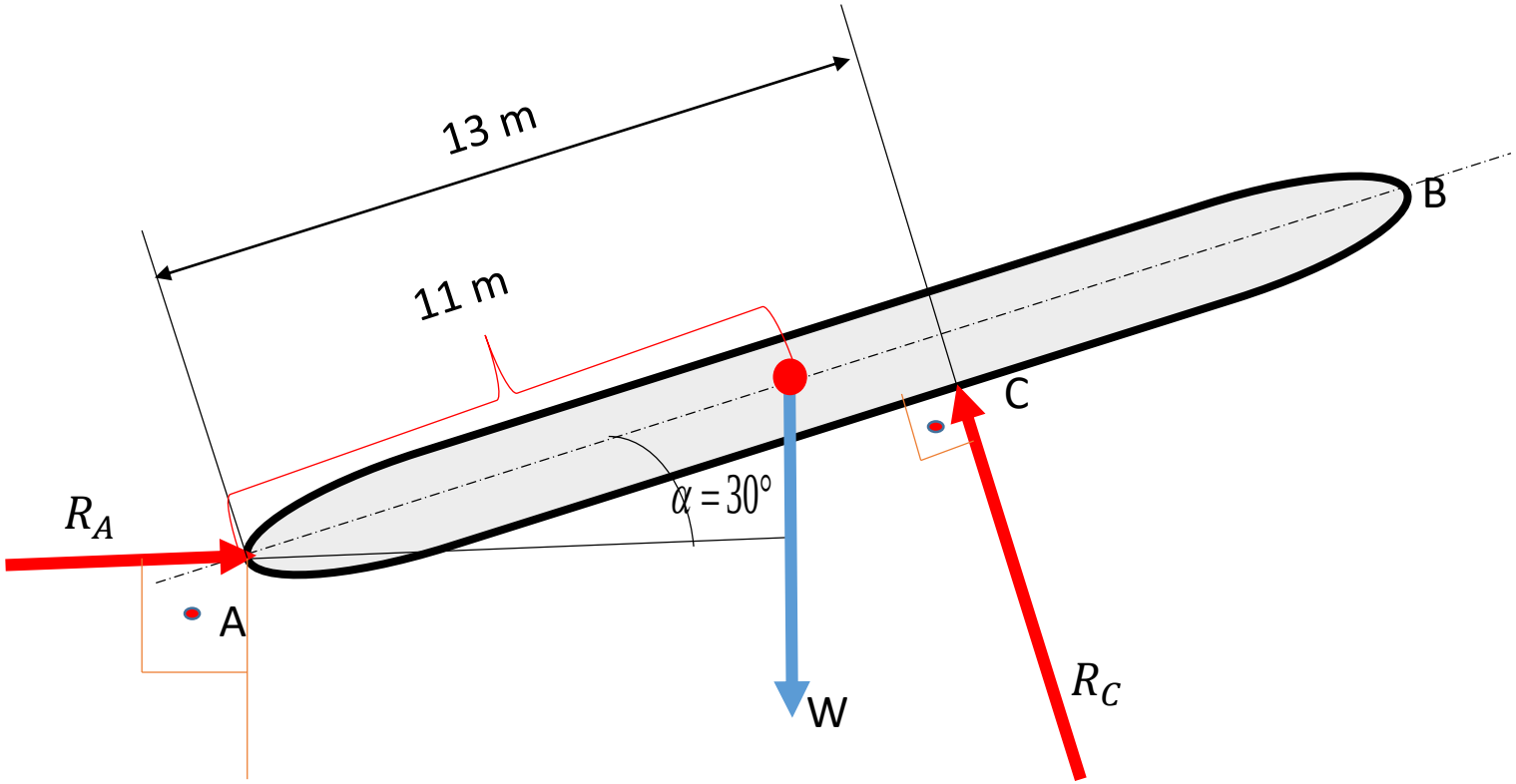
Yatay ile α=30° lik açı yapan, 22 m uzunluğunda ve 50 kN ağırlığında üniform kalas şekilde görüldüğü gibi *A* ve *C* de sürtünmesiz temas etmekte dengede durmaktadır. *A* ve *C* deki mesnet tepkilerini hesaplayınız.

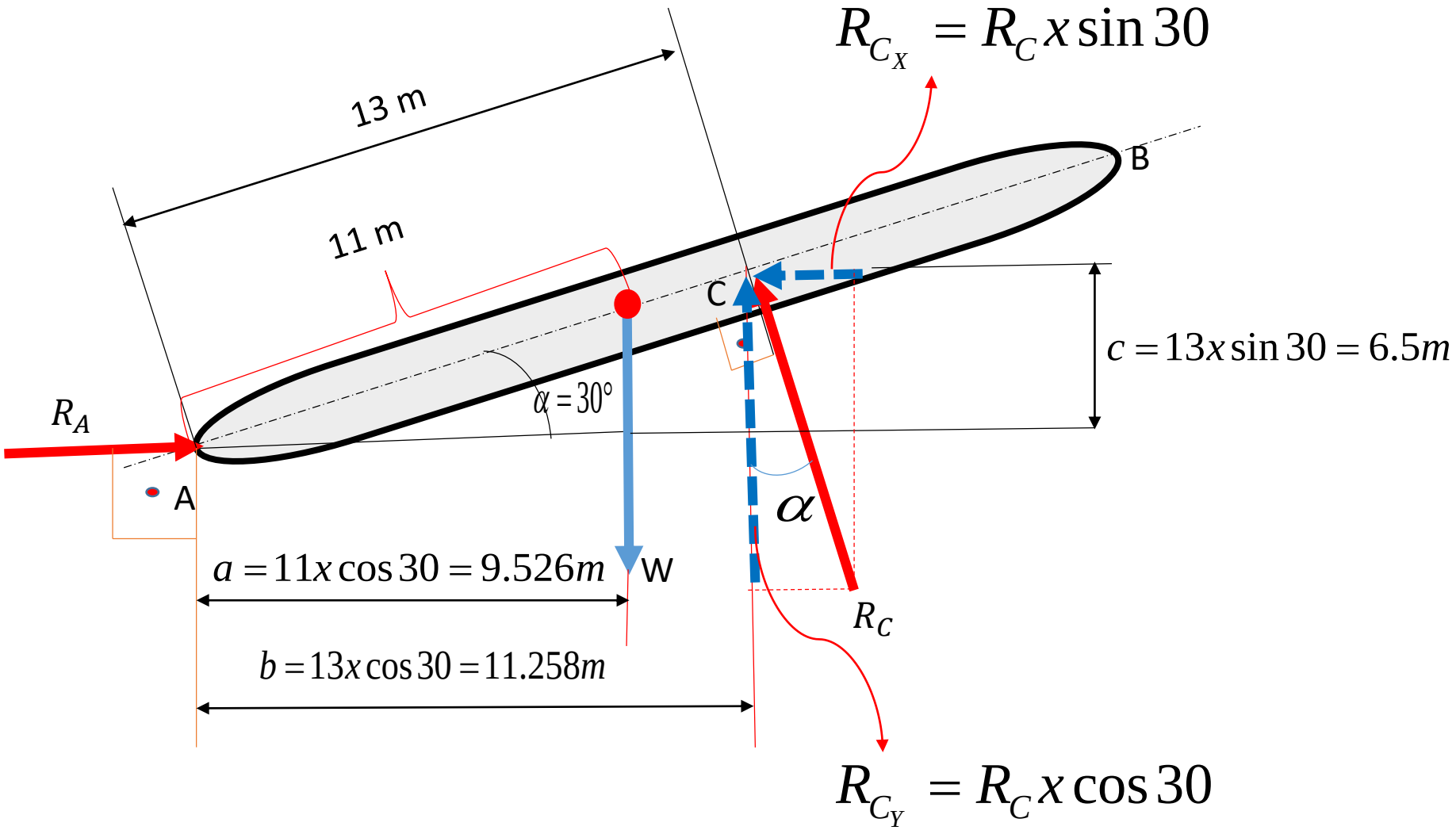


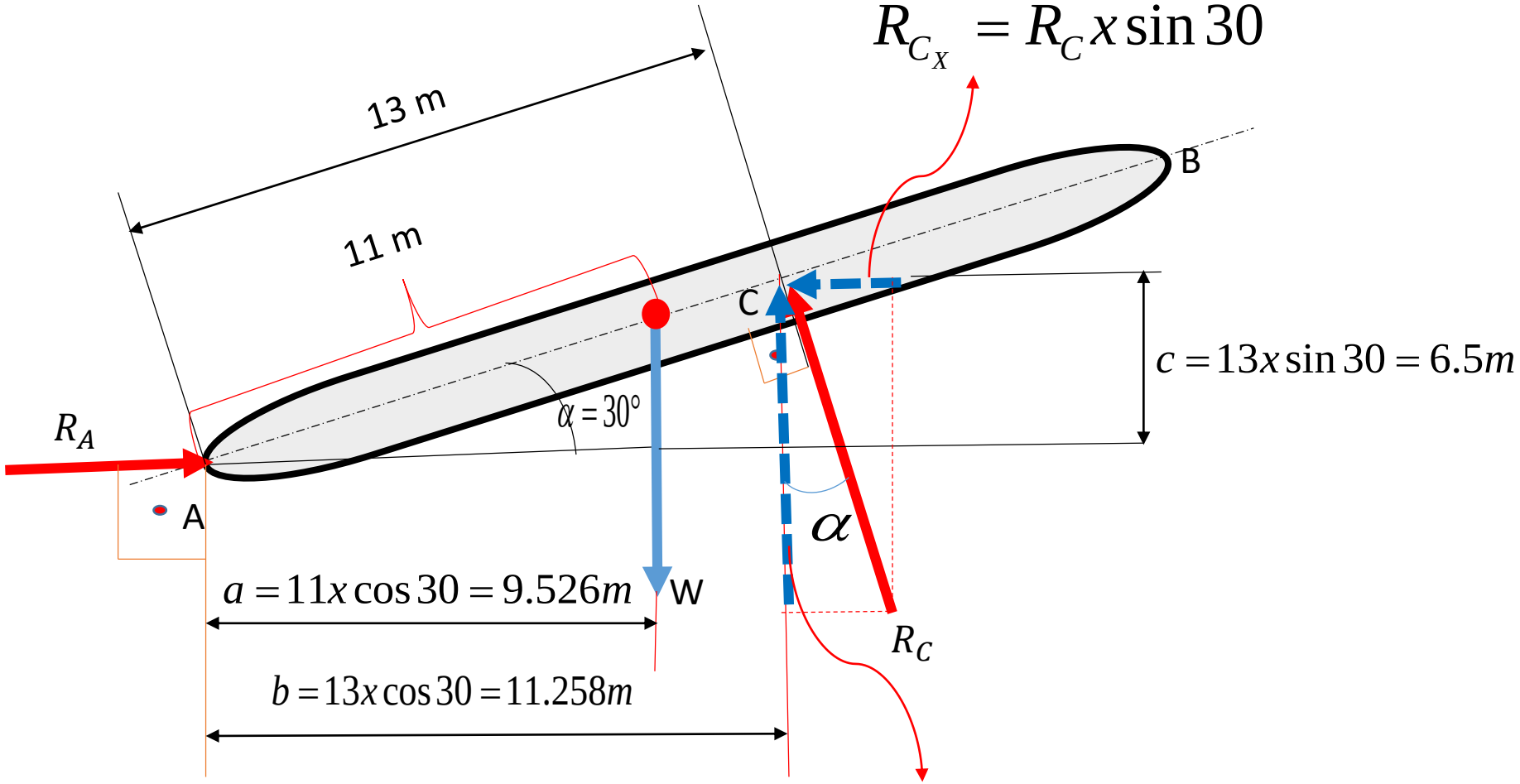

13 m
B
C
α
A
W


13 m
B
C
$\alpha$
$R_A$
A
W


13 m
B
C
$\alpha$
$R_A$
A
W
$R_C$


13 m
11 m
B
C
$R_A$
$\alpha = 30°$
A
W
$R_C$


13 m
11 m
B
C
A
$R_A$
$\alpha = 30°$
$c = 13x\sin 30 = 6.5m$
$\alpha$
$a = 11x\cos 30 = 9.526m$
W
$R_C$
$b = 13x\cos 30 = 11.258m$
$R_{C_X} = R_C x\sin 30$
$R_{C_Y} = R_C x\cos 30$




$$R_{C_X} = R_C x \sin 30$$

$$R_{C_Y} = R_C x \cos 30$$

$$+\sum M_A = 0$$

$$-Wxa + R_{C_Y} xb + R_{C_X} xc = 0$$

$$-50x9.526 + R_C x\cos 30x11.258 + R_C x\sin 30x6.5 = 0$$

$$R_C = 36.639kN$$

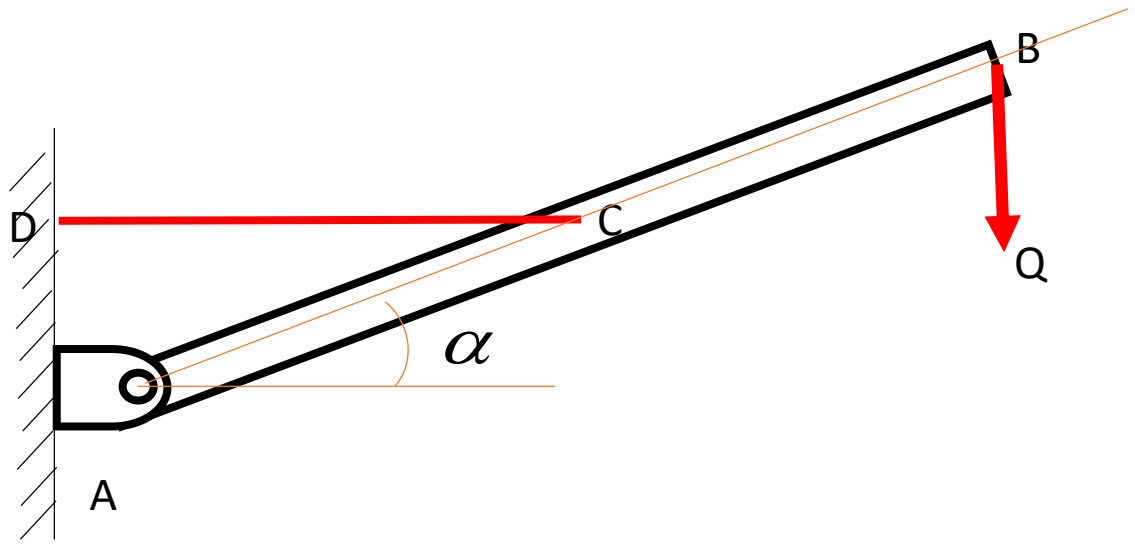
$$\sum F_x = 0 \quad \pm$$

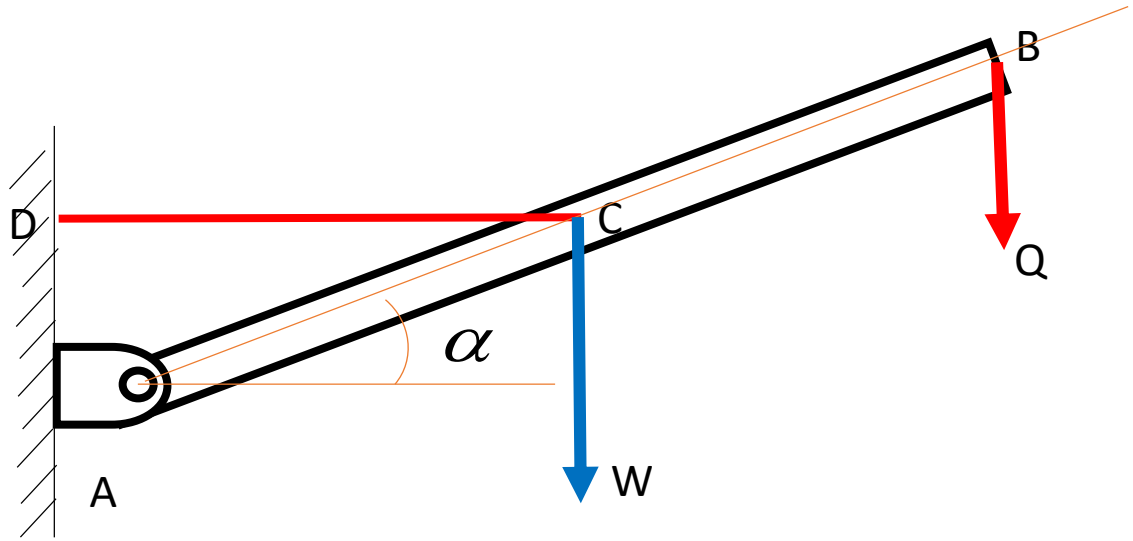
$$R_A - R_{C_X} = o$$

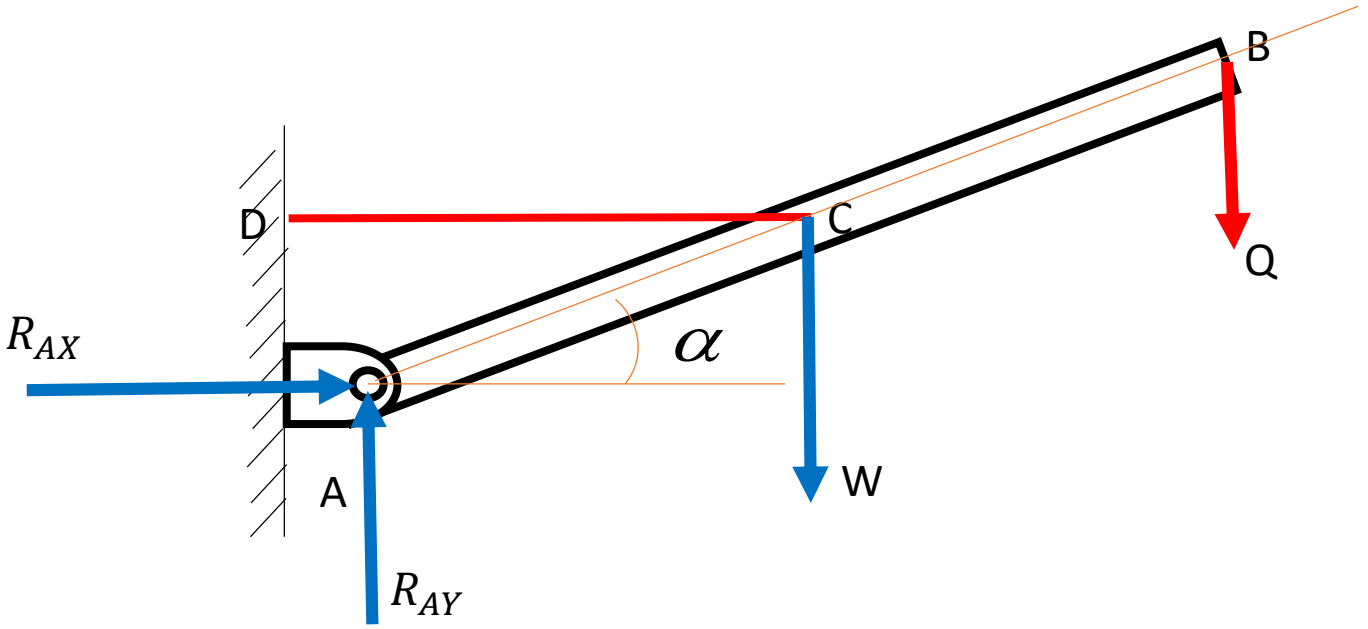
$$R_A = R_{C_X} = R_C x \sin 30$$

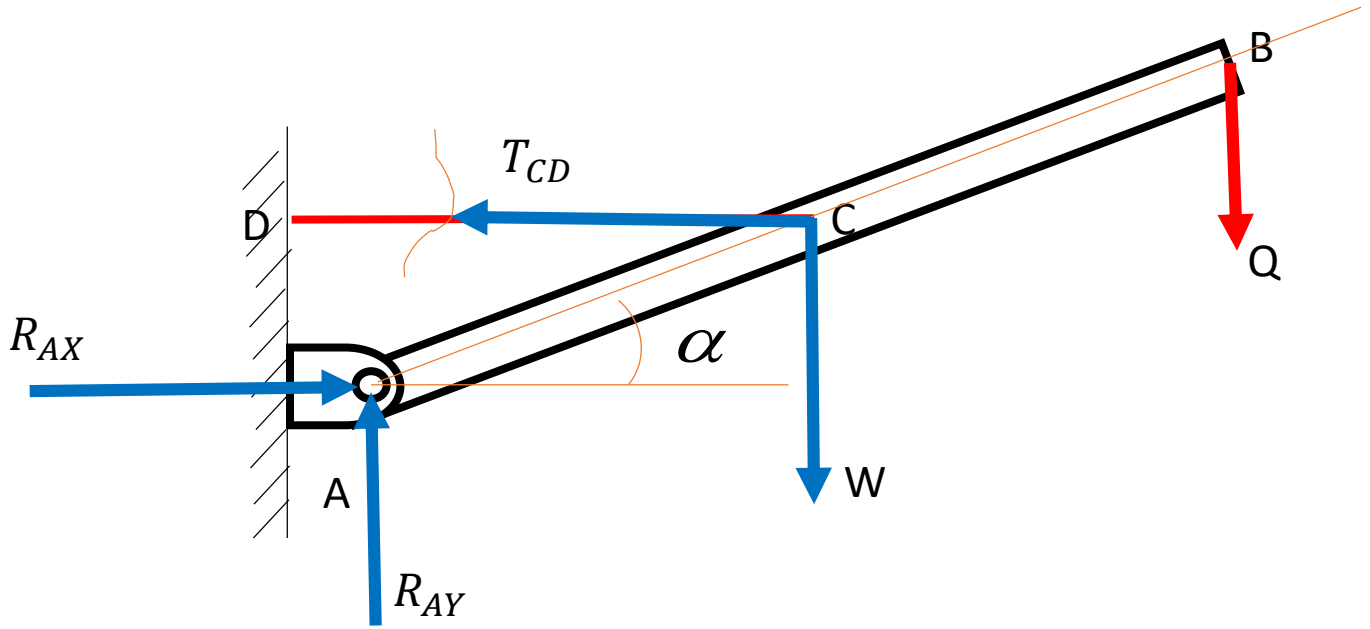
$$R_A = 36.639x\sin 30 = 18.319kN$$

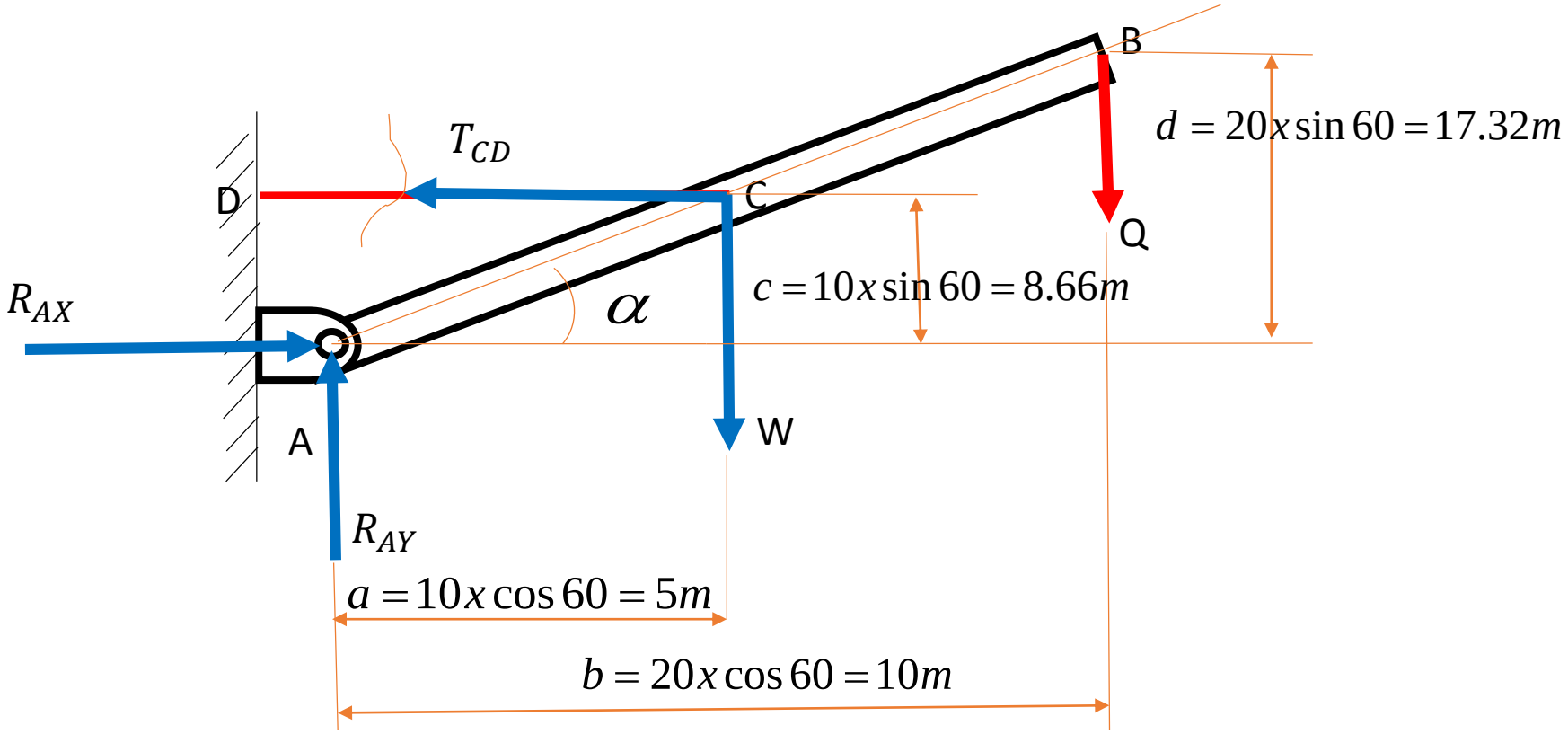
Yatay ile α=60° lik açı yapan, 20 m uzunluğunda ve 60 kN ağırlığında üniform kalas şekilde görüldüğü gibi A noktasında sabit mafsallı mesnet ile bağlanmış ve B ucunda da Q=40 kN luk yükü taşımakta, AB kalasının orta noktasına bağlanmış yatay CD halatı ile dengede tutulmaktadır. A mesnet tepkisini ve CD halatındaki germe kuvvetini hesaplayınız.

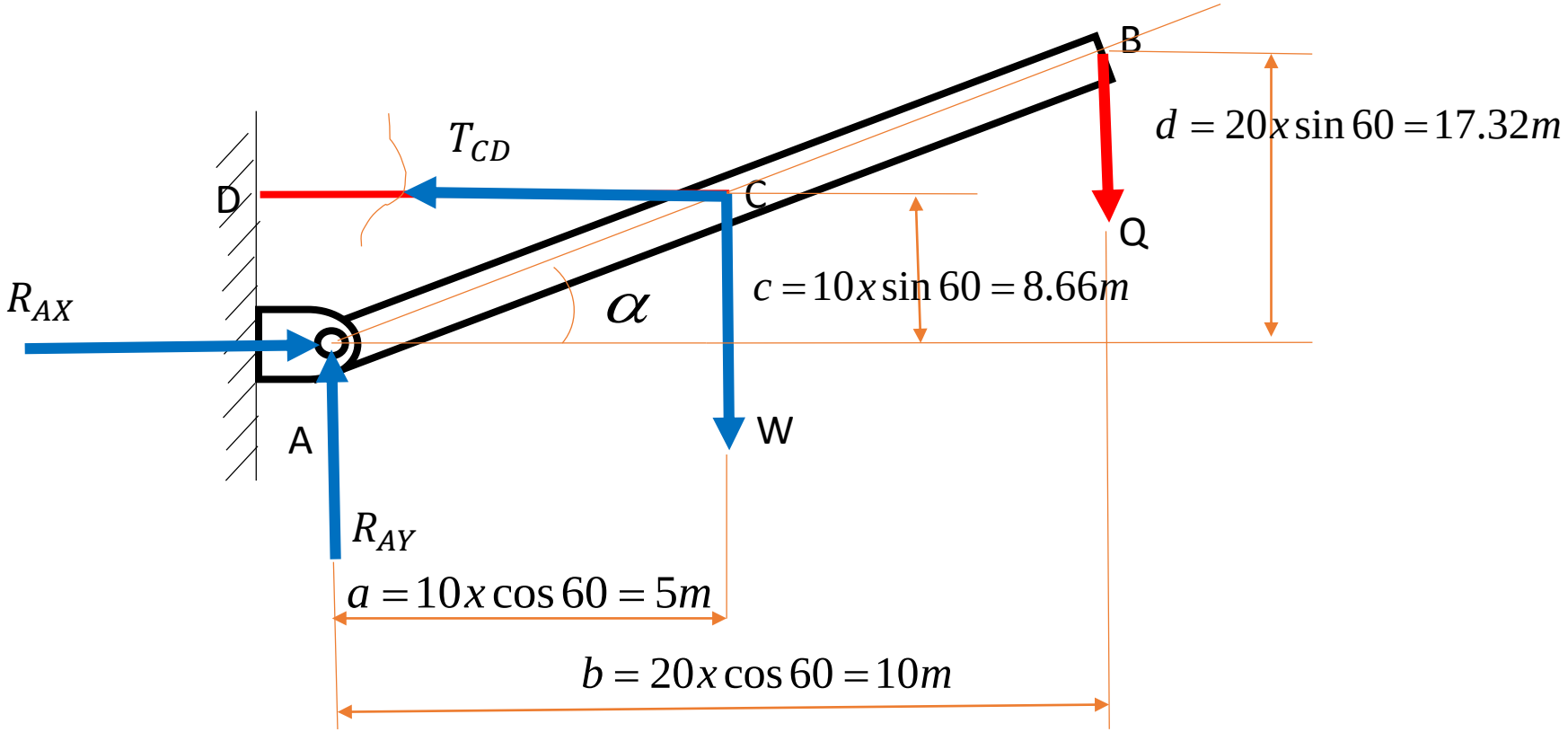



B
D
C
Q
α
A
W


B
D
C
Q
$R_{AX}$
$\alpha$
A
W
$R_{AY}$


B
$T_{CD}$
D
C
Q
$R_{AX}$
$\alpha$
W
A
$R_{AY}$


B
$d = 20x\sin 60 = 17.32m$
$T_{CD}$
D
C
Q
$R_{AX}$
$\alpha$
$c = 10x\sin 60 = 8.66m$
A
W
$R_{AY}$
$a = 10x\cos 60 = 5m$
$b = 20x\cos 60 = 10m$

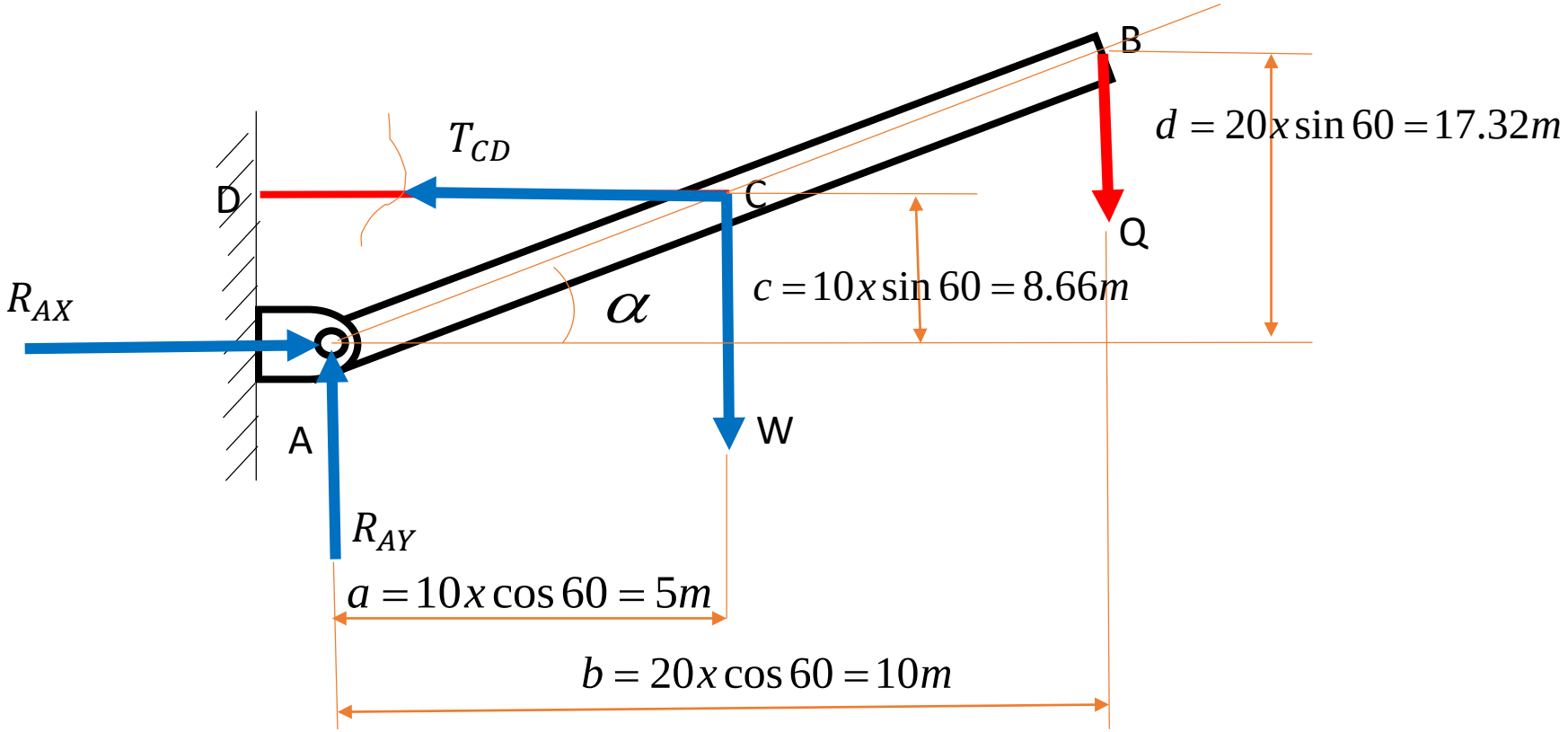



$+\circlearrowleft \sum M_A = 0$

$$-Wxa + T_{CD}xc - Qxb = 0$$

$$-60x5 + T_{CD}x8.66 - 40x10 = 0$$

$$T_{CD} = 80.831kN$$




$$+\circlearrowleft \sum M_A = 0$$

$$-Wxa + T_{CD}xc - Qxb = 0$$

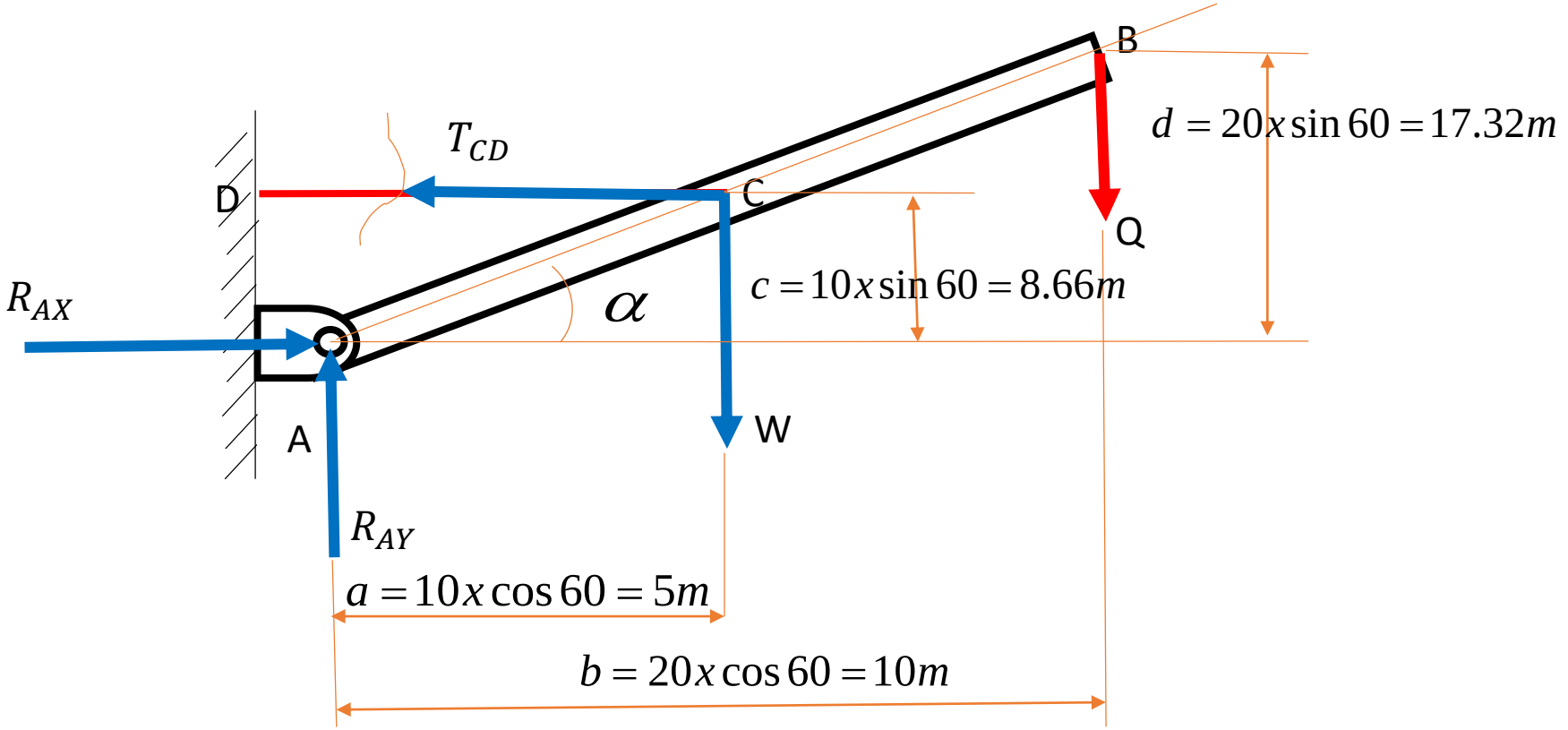
$$-60x5 + T_{CD}x8.66 - 40x10 = 0$$

$$T_{CD} = 80.831kN$$

$$\sum F_x = 0 \quad \pm$$

$$R_{AX} - T_{CD} = o$$

$$R_{AX} = T_{CD} = 80.831kN$$




$\sum M_A = 0$ (+)

$-Wxa + T_{CD}xc - Qxb = 0$

$-60x5 + T_{CD}x8.66 - 40x10 = 0$

$T_{CD} = 80.831kN$

$\sum F_x = 0$ (±)

$R_{AX} - T_{CD} = o$

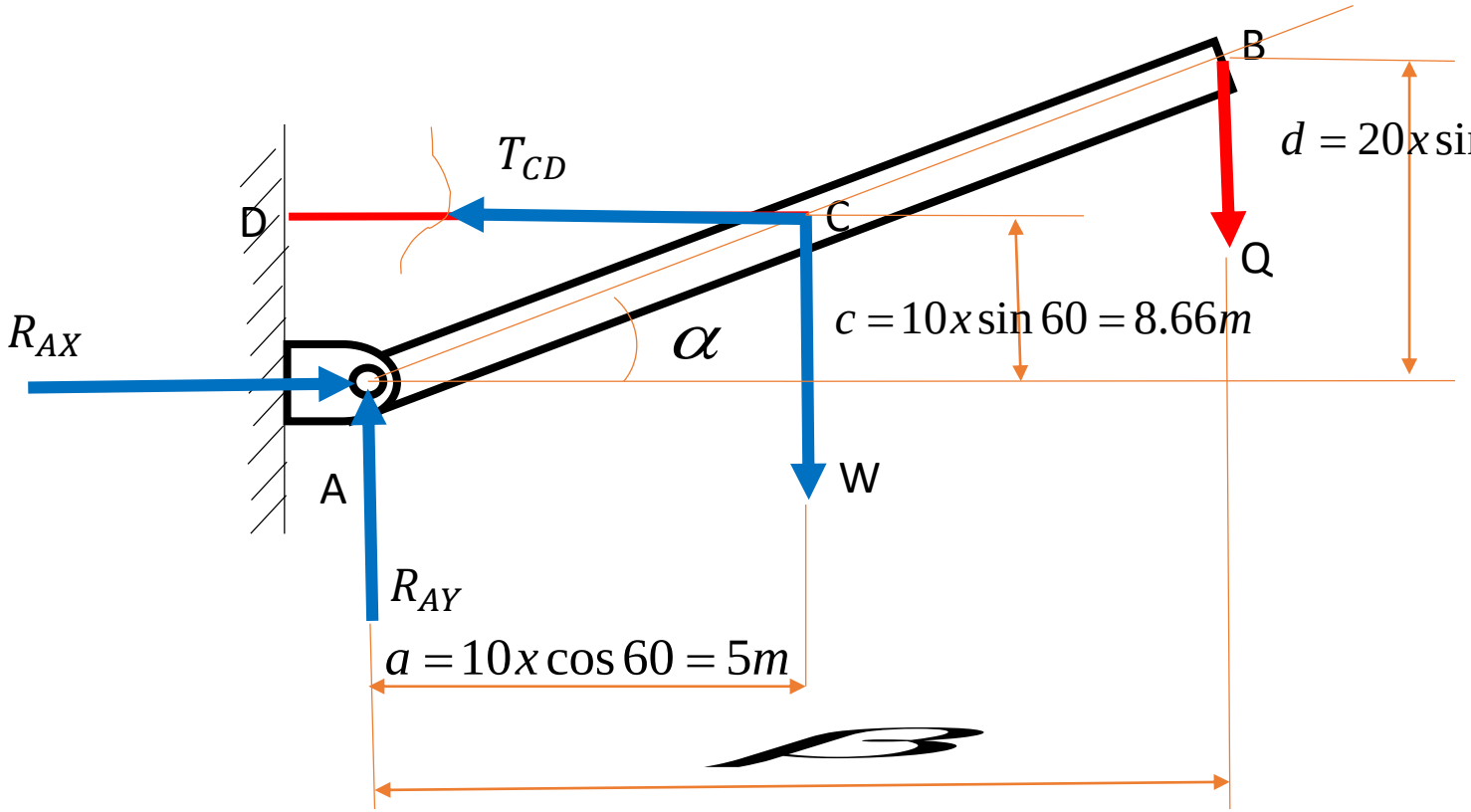
$R_{AX} = T_{CD} = 80.831kN$

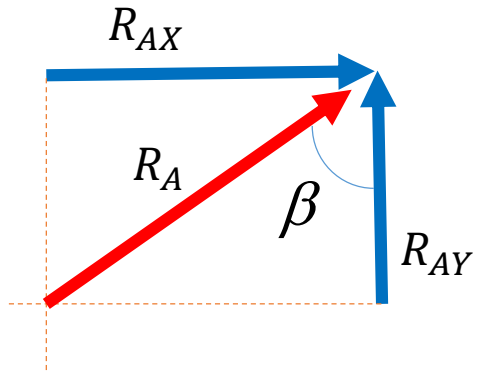
$\sum F_y = 0$ (+↑ −↓)

$R_{AY} - W - Q = 0$

$R_{AY} = W + Q = 60 + 40 = 100kN$

$R_{AY} = 100kN$



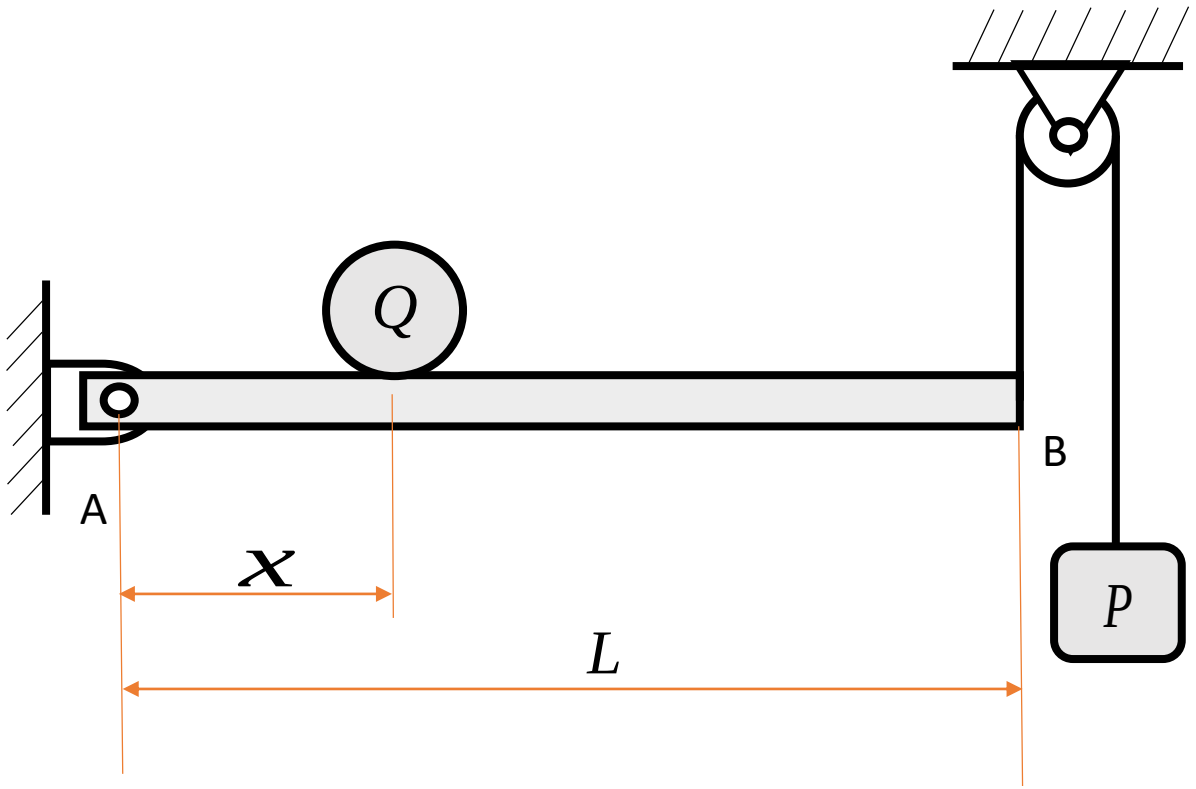



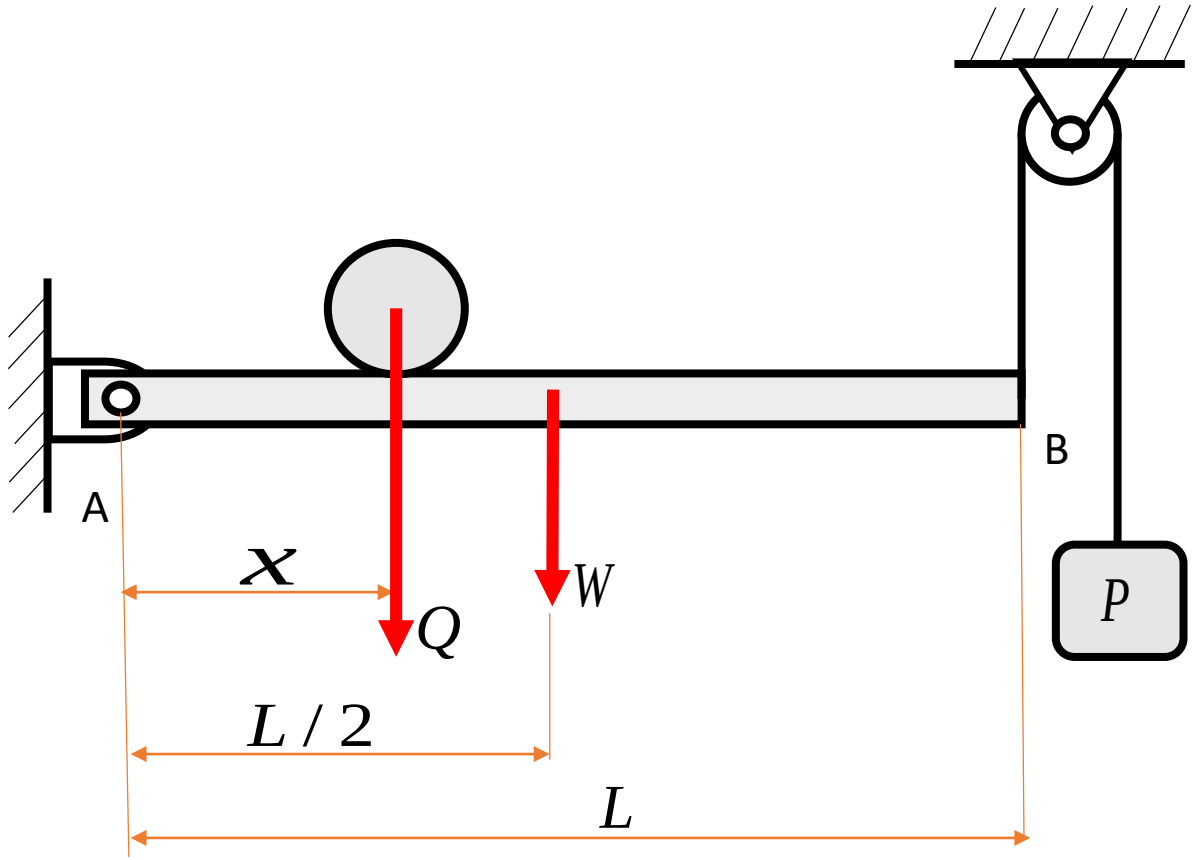

$$R_A = \sqrt{R_{AX}^2 + R_{AY}^2} = \sqrt{80.831^2 + 100^2} = 128.583kN$$

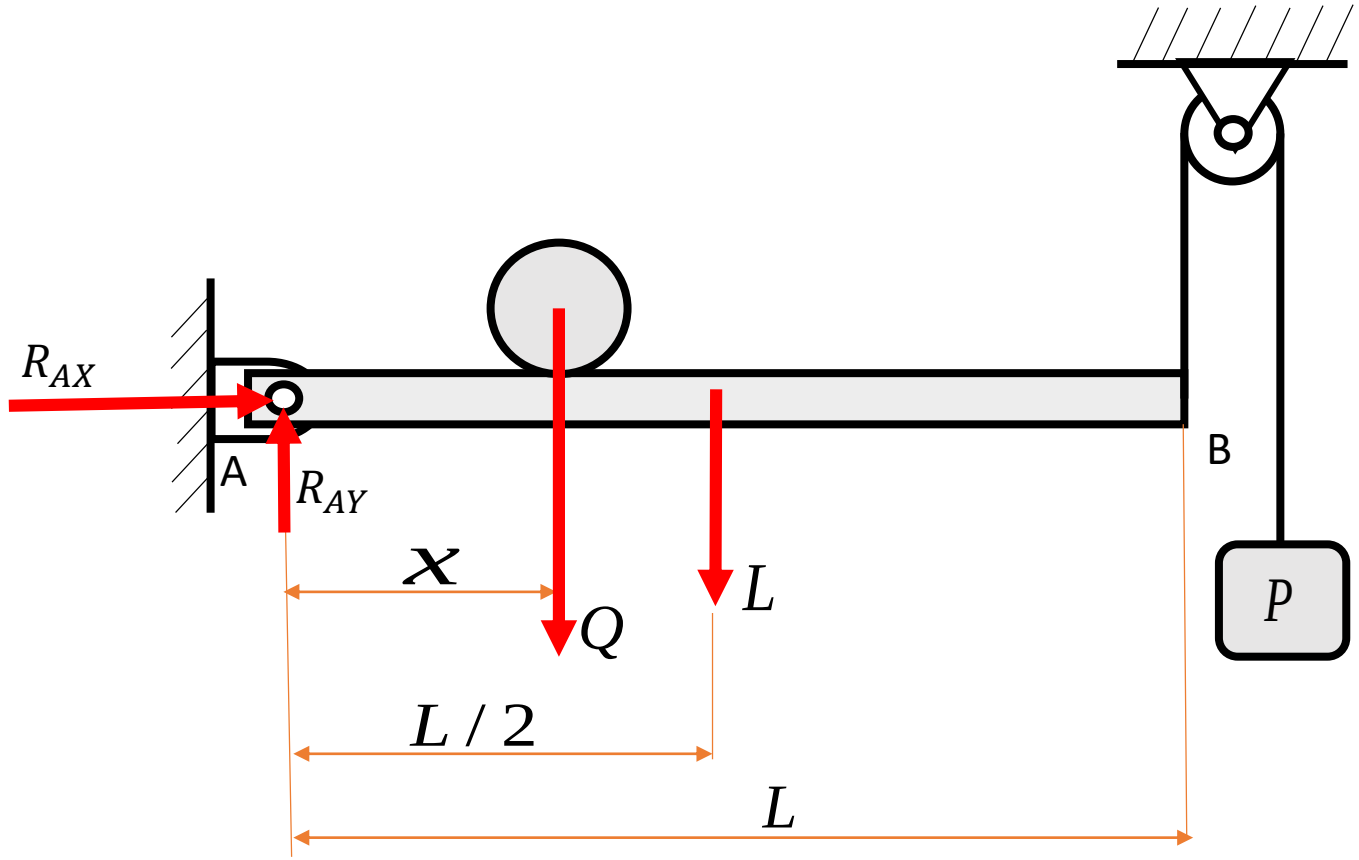
$$\tan\beta = \frac{R_{AX}}{R_{AY}} = \frac{80.831}{100} = 0.808$$

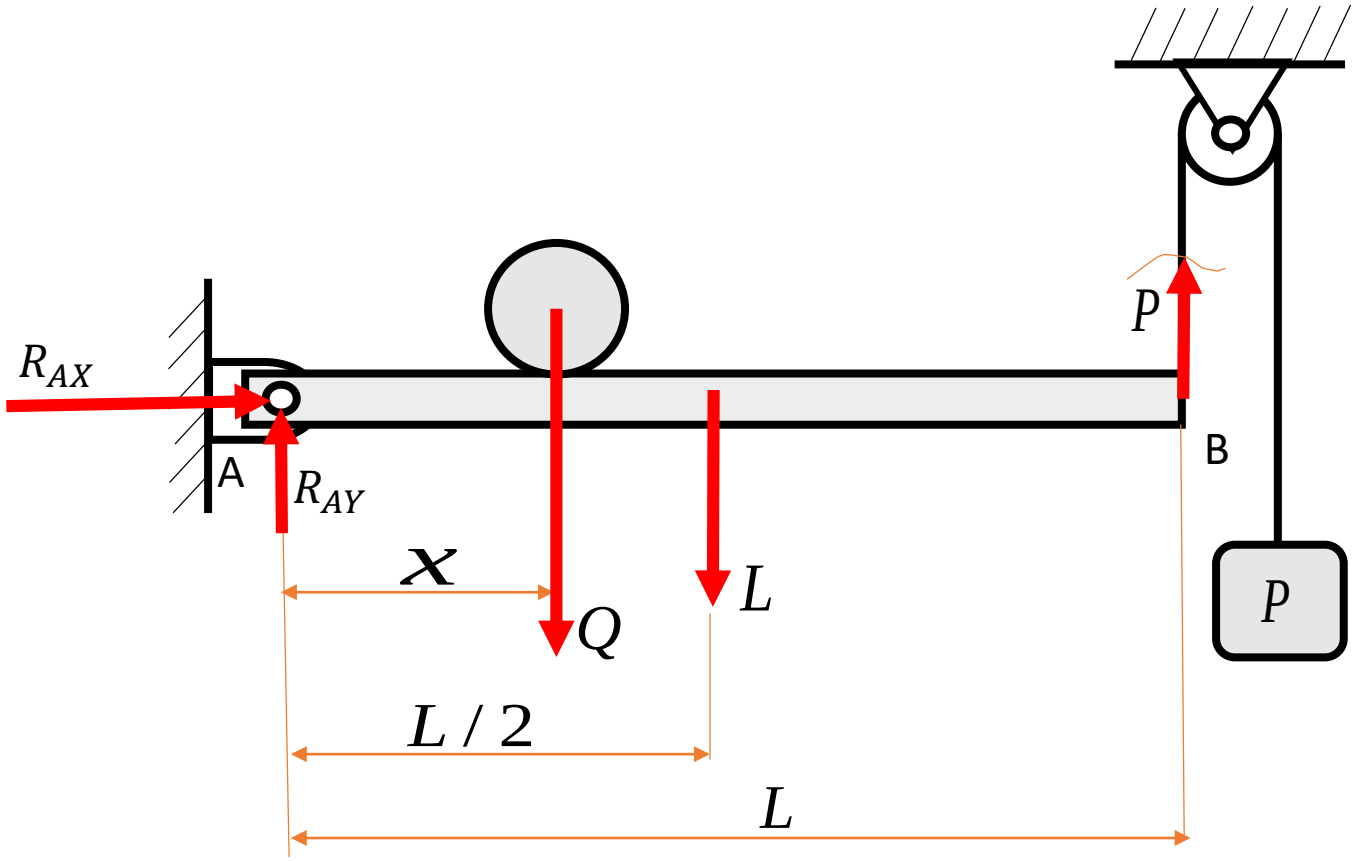
$$\beta = 38.948°$$

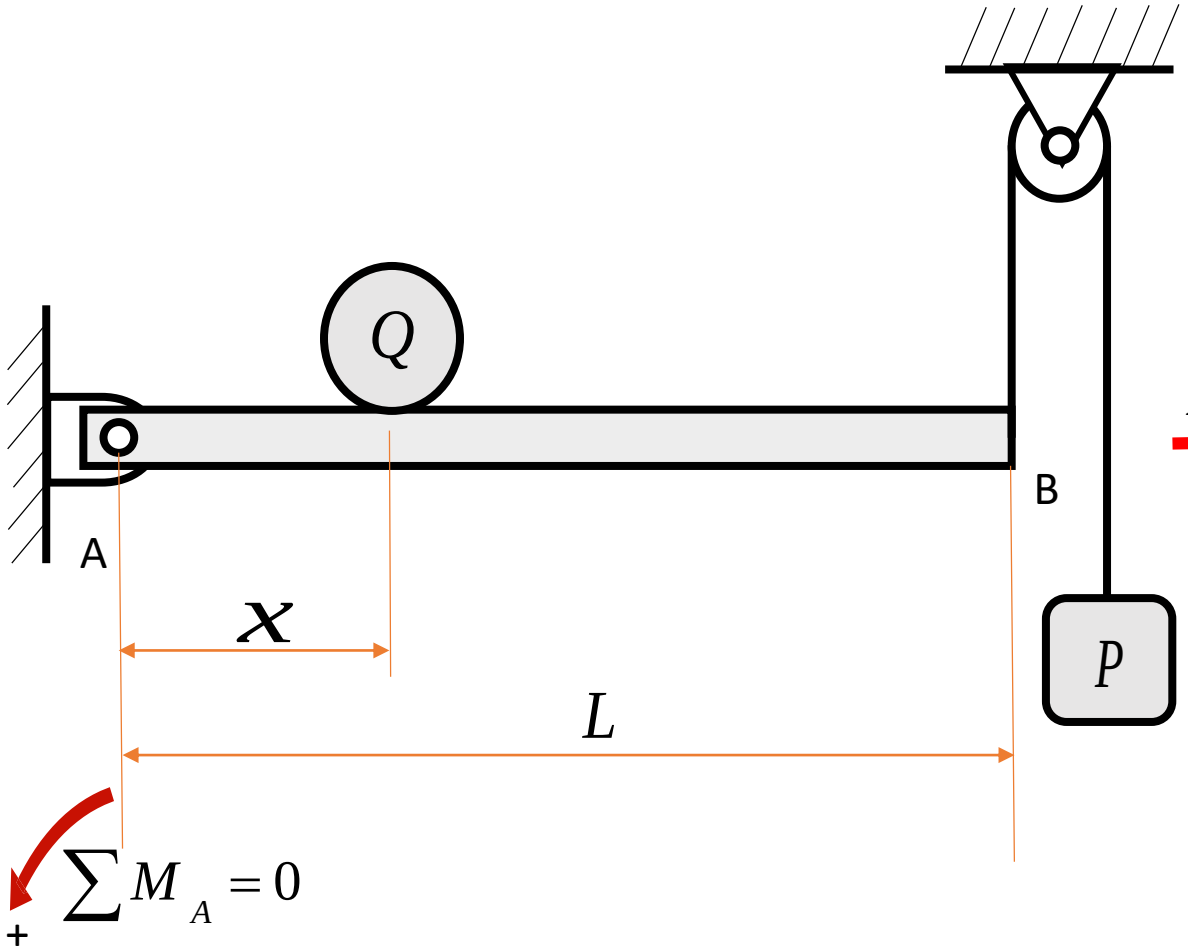
Şekilde L=10m uzunluğunda 90 kN ağırlığında üniform AB çubuğu A noktasında sabit mafsallı mesnet ile bağlanmış ve B ucunda da P=120 kN luk yükü taşımakta, AB çubuğunun yatay olarak dengede olabilmesi için Q=100 kN luk ağırlık çubuğun A ucundan ne kadar X=? mesafeye yerleştirilmelidir? A mesnet tepkisini hesaplayınız.

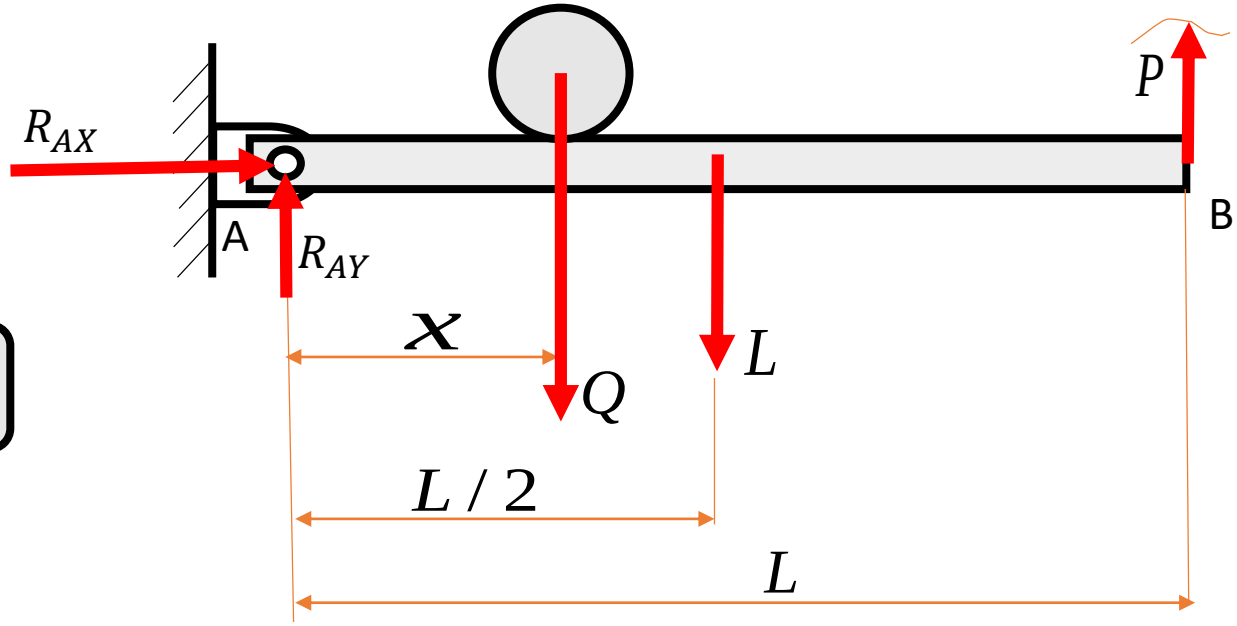



A
B
x
Q
W
P
L / 2
L


$R_{AX}$
A
$R_{AY}$
$x$
$Q$
$L$
$L / 2$
$L$
B
$P$


$R_{AX}$
A
$R_{AY}$
$x$
$Q$
$L$
$L / 2$
$L$
$P$
B
$P$

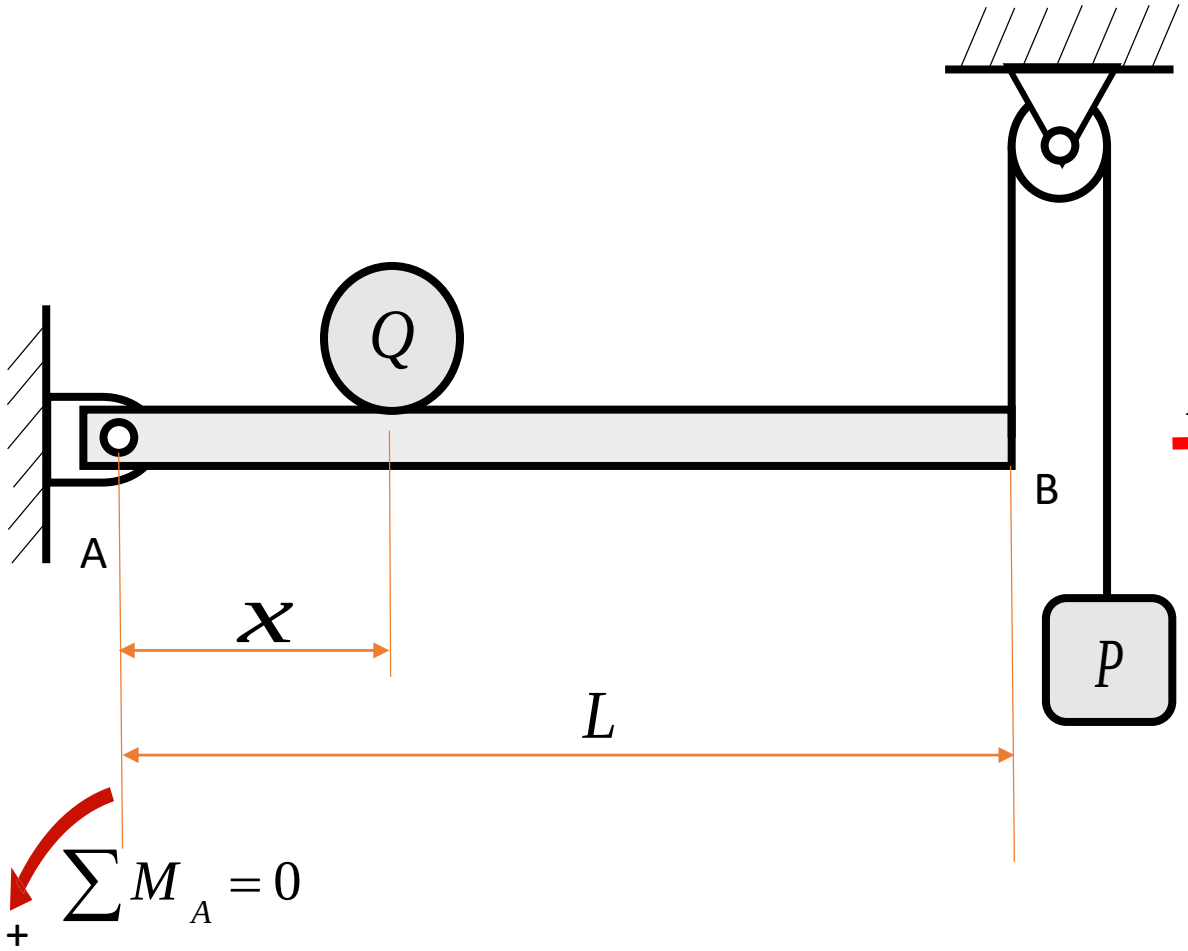


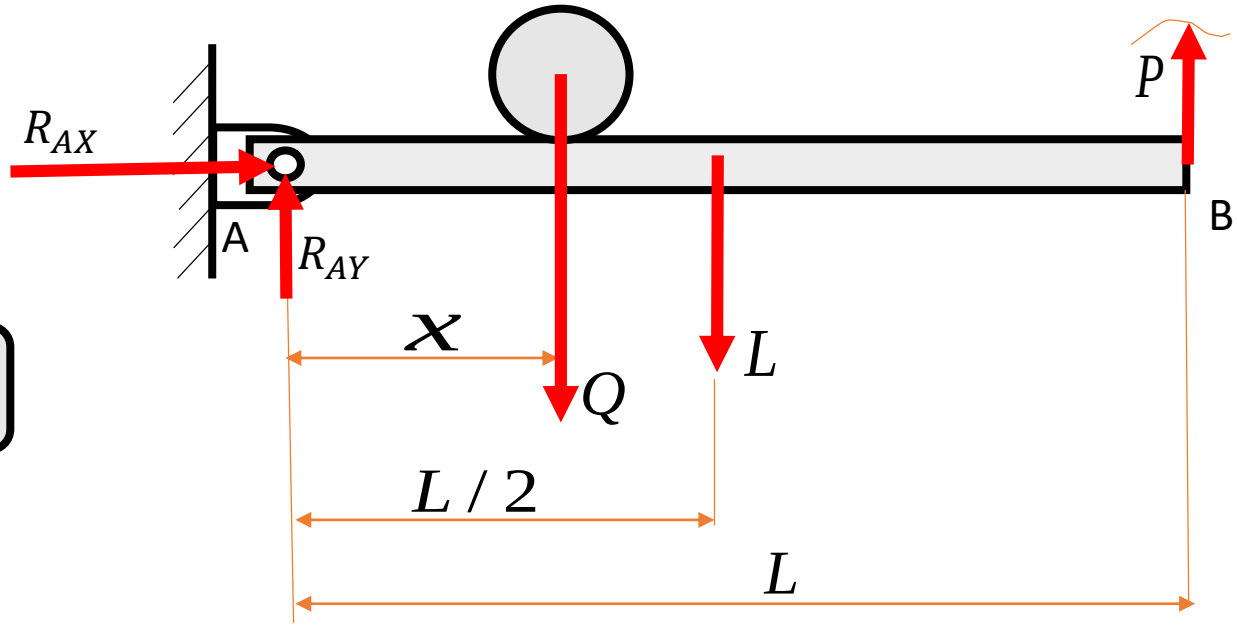




$$QxX + WxL - Px10 = 0$$

$$100xX + 90x5 - 120x10 = 0$$

$$X = 7.5m$$



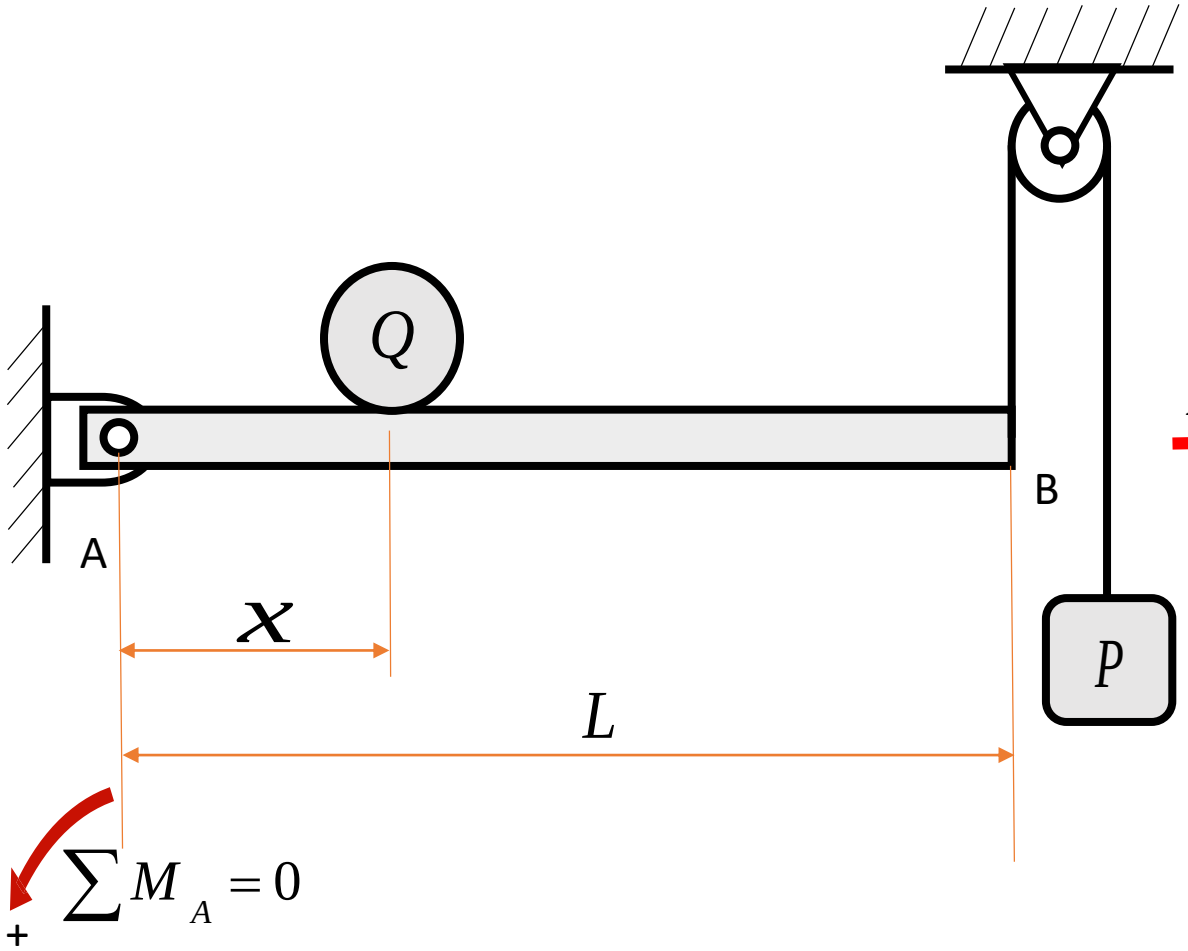

$$QxX + WxL - Px10 = 0$$

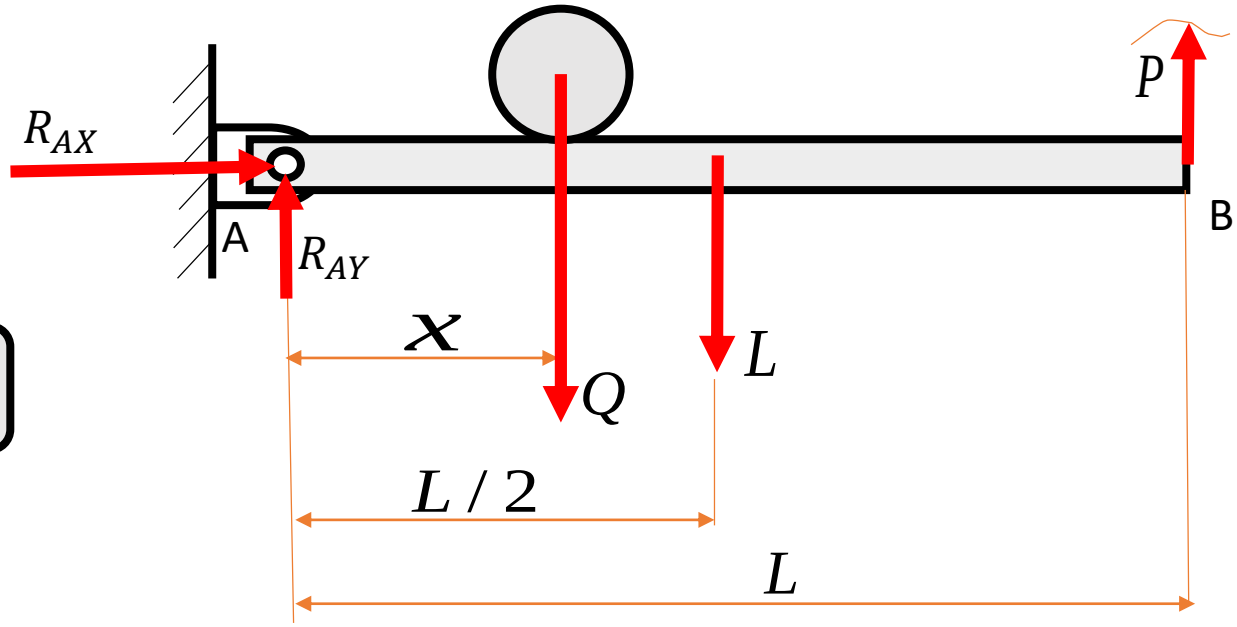
$$100xX + 90x5 - 120x10 = 0$$

$$X = 7.5m$$

$$\sum F_x = 0 \square \quad \pm$$

$$R_{AX} = o$$




$$QxX + WxL - Px10 = 0$$

$$100xX + 90x5 - 120x10 = 0$$

$$X = 7.5m$$

$$\sum F_x = 0 \quad \pm$$

$$R_{AX} = o$$

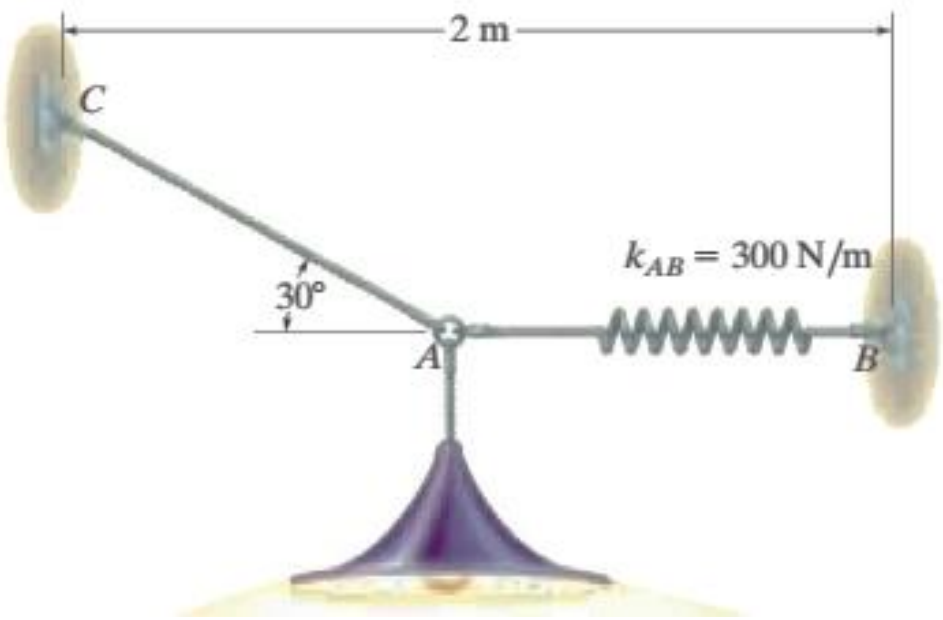
$$\sum F_y = 0 \overset{+}{\uparrow\downarrow}_{-}$$

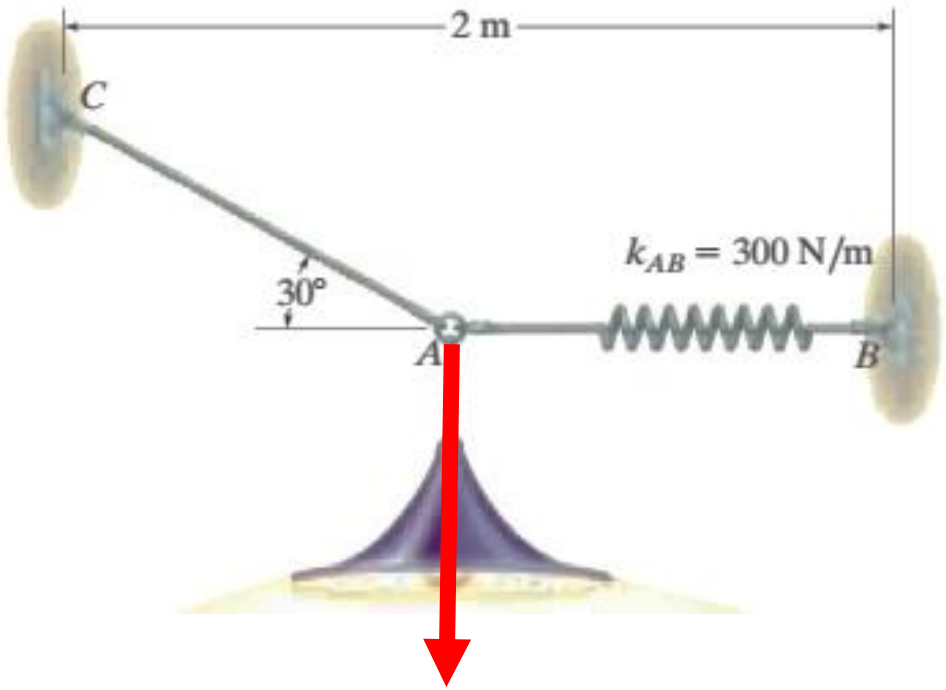
$$R_{AY} - W - Q + P = 0$$

$$R_{AY} = W + Q - P = 90 + 100 - 120$$

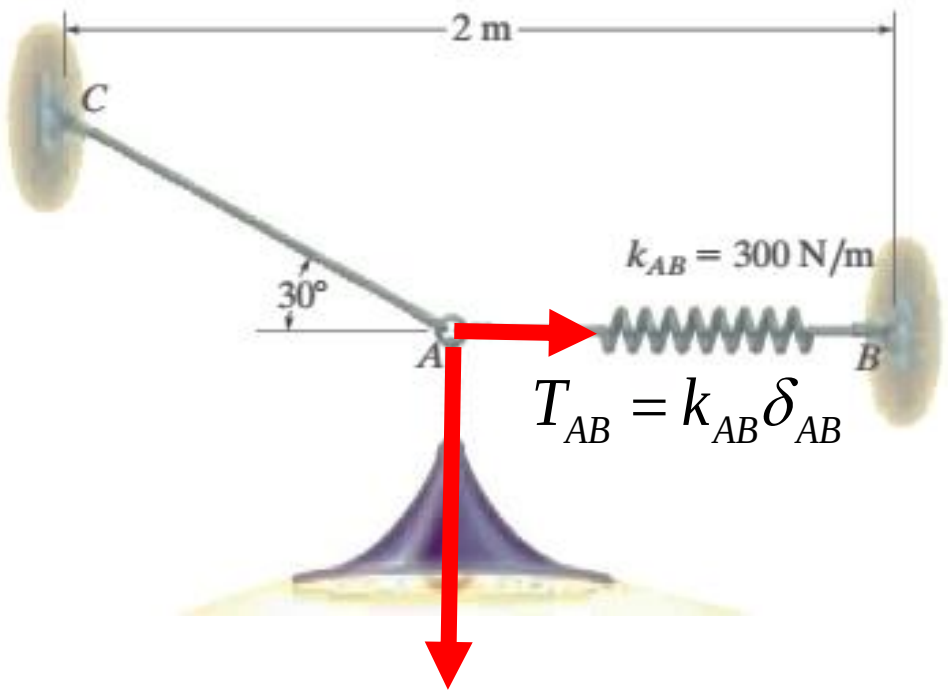
$$R_{AY} = 70kN$$

Şekildeki 8 kg kütleye sahip lambayı AC halatı ve yatay olarak gerilmiş AB yayı taşımaktadır. AB yayının uzamamış boyu 0.4 m ve yay sabiti 300 N/m dir. AC halatının uzunluğunu hesaplayınız.

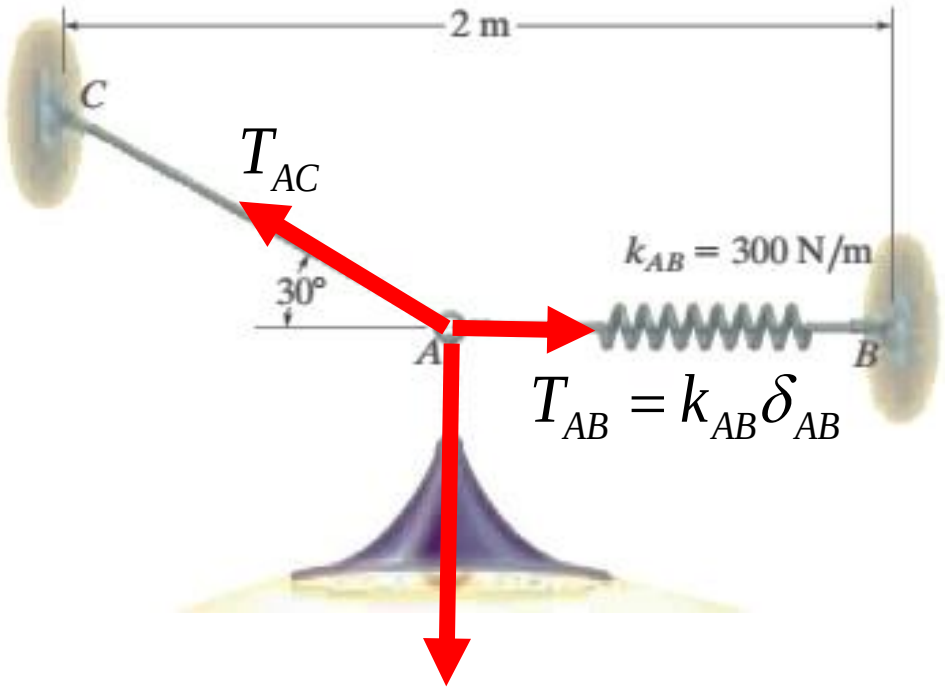




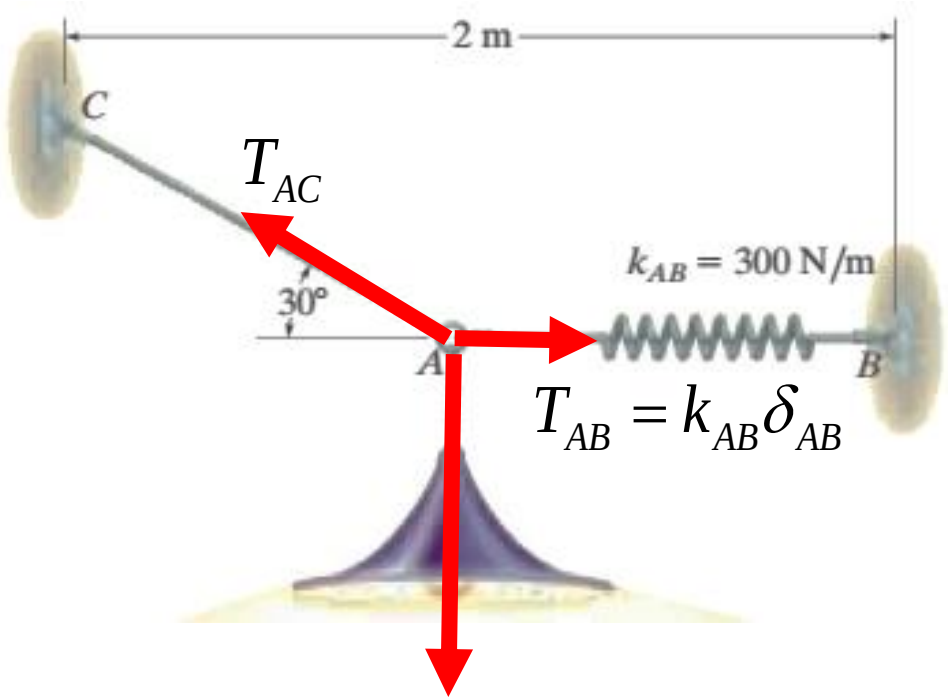

$$W = mg = 8x9.81 = 78.5N$$




$$W = mg = 8x9.81 = 78.5N$$




$$W = mg = 8x9.81 = 78.5N$$




$$\stackrel{+}{\rightarrow} \Sigma F_x = 0; \qquad T_{AB} - T_{AC}\cos 30° = 0$$

$$+\uparrow \Sigma F_y = 0; \qquad T_{AC}\sin 30° - 78.5\ \text{N} = 0$$

$$T_{AC} = 157.0\ \text{N}$$

$$T_{AB} = 135.9\ \text{N}$$

$$T_{AB} = k_{AB}s_{AB}; \qquad 135.9\ \text{N} = 300\ \text{N/m}(s_{AB})$$

$$s_{AB} = 0.453\ \text{m}$$

$$l_{AB} = l'_{AB} + s_{AB}$$

$$l_{AB} = 0.4\ \text{m} + 0.453\ \text{m} = 0.853\ \text{m}$$

$$2\ \text{m} = l_{AC}\cos 30° + 0.853\ \text{m}$$

$$l_{AC} = 1.32\ \text{m}$$